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### Microelectronic assemblies

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#### Abstract

Microelectronic assemblies, and related devices and methods, are disclosed herein. For example, in some embodiments, a microelectronic assembly may include a package substrate having a first surface and an opposing second surface, and a die secured to the package substrate, wherein the die has a first surface and an opposing second surface, the die has first conductive contacts at the first surface and second conductive contacts at the second surface, and the first conductive contacts are coupled to conductive pathways in the package substrate by first non-solder interconnects.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of (and claims the benefit of priority under 35 U.S.C. § 120 to) U.S. application Ser. No. 17/129,134, filed on Dec. 21, 2020 and entitled “MICROELECTRONIC ASSEMBLIES,” which application is a continuation of U.S. application Ser. No. 16/650,499 filed on Mar. 25, 2020 and entitled “MICROELECTRONIC ASSEMBLIES,” which is a national stage application under 35 U.S.C. § 371 of PCT International Application Serial No. PCT/US2017/068921, filed on Dec. 29, 2017 and entitled “MICROELECTRONIC ASSEMBLIES,” which are hereby incorporated by reference herein in their entireties.

### BACKGROUND

(1) Integrated circuit dies are conventionally coupled to a package substrate for mechanical stability and to facilitate connection to other components, such as circuit boards. The interconnect pitch achievable by conventional substrates is constrained by manufacturing, materials, and thermal considerations, among others.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments will be readily understood by the following detailed description in conjunction with the accompanying drawings. To facilitate this description, like reference numerals designate like structural elements. Embodiments are illustrated by way of example, not by way of limitation, in the figures of the accompanying drawings.
- (2) FIG. 1 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments.
- (3) FIG. 2 is a bottom view of a die included in the microelectronic assembly of FIG. 1, in accordance with various embodiments.
- (4) FIGS. 3-11 are side, cross-sectional views of example microelectronic assemblies, in accordance with various embodiments.
- (5) FIGS. 12-16 are top views of example arrangements of multiple dies in a microelectronic assembly, in accordance with various embodiments.
- (6) FIGS. 17A-17F are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly of FIG. 5, in accordance with various embodiments.
- (7) FIGS. 18A-18B are side, cross-sectional views of various stages in another example process for manufacturing the microelectronic assembly of FIG. 5, in accordance with various embodiments.
- (8) FIGS. 19A-19H are side, cross-sectional views of various stages in another example process for manufacturing the microelectronic assembly of FIG. 5, in accordance with various embodiments.
- (9) FIGS. 20-22 are side, cross-sectional views of example microelectronic assemblies, in accordance with various embodiments.
- (10) FIGS. 23A-23B are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly of FIG. 20, in accordance with various embodiments.
- (11) FIGS. 24A-24E are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly of FIG. 21, in accordance with various embodiments.

- (12) FIGS. 25A-25F are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly of FIG. 22, in accordance with various embodiments.
- (13) FIGS. 26A-26D are side, cross-sectional views of various stages in another example process for manufacturing the microelectronic assembly of FIG. 21, in accordance with various embodiments.
- (14) FIG. 27 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments.
- (15) FIG. 28 is a top view of a wafer and dies that may be included in a microelectronic assembly, in accordance with any of the embodiments disclosed herein.
- (16) FIG. 29 is a cross-sectional side view of an integrated circuit (IC) device that may be included in a microelectronic assembly, in accordance with any of the embodiments disclosed herein.
- (17) FIG. 30 is a cross-sectional side view of an IC device assembly that may include a microelectronic assembly, in accordance with any of the embodiments disclosed herein.
- (18) FIG. 31 is a block diagram of an example electrical device that may include a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

#### DETAILED DESCRIPTION

- (19) Microelectronic assemblies, and related devices and methods, are disclosed herein. For example, in some embodiments, a microelectronic assembly may include a package substrate having a first surface and an opposing second surface and a die secured to the package substrate, wherein the die has a first surface and an opposing second surface, the die has first conductive contacts at the first surface and second conductive contacts at the second surface, and the first conductive contacts are coupled to conductive pathways in the package substrate by first non-solder interconnects.
- (20) Communicating large numbers of signals between two or more dies in a multi-die integrated circuit (IC) package is challenging due to the increasingly small size of such dies, thermal constraints, and power delivery constraints, among others. Various ones of the embodiments disclosed herein may help achieve reliable attachment of multiple IC dies at a lower cost, with improved power efficiency, with higher bandwidth, and/or with greater design flexibility, relative to conventional approaches. Various ones of the microelectronic assemblies disclosed herein may exhibit better power delivery and signal speed while reducing the size of the package relative to conventional approaches. The microelectronic assemblies disclosed herein may be particularly advantageous for small and low-profile applications in computers, tablets, industrial robots, and consumer electronics (e.g., wearable devices).
- (21) In the following detailed description, reference is made to the accompanying drawings that form a part hereof wherein like numerals designate like parts throughout, and in which is shown, by way of illustration, embodiments that may be practiced. It is to be understood that other embodiments may be utilized, and structural or logical changes may be made without departing from the scope of the present disclosure. Therefore, the following detailed description is not to be taken in a limiting sense.
- (22) Various operations may be described as multiple discrete actions or operations in turn, in a manner that is most helpful in understanding the claimed subject matter. However, the order of description should not be construed as to imply that these operations are necessarily order dependent. In particular, these operations may not be performed in the order of presentation. Operations described may be performed in a different order from the described embodiment. Various additional operations may be performed, and/or described operations may be omitted in additional embodiments.
- (23) For the purposes of the present disclosure, the phrase “A and/or B” means (A), (B), or (A and B). For the purposes of the present disclosure, the phrase “A, B, and/or C” means (A), (B), (C), (A and B), (A and C), (B and C), or (A, B, and C). The drawings are not necessarily to scale. Although many of the drawings illustrate rectilinear structures with flat walls and right-angle corners, this is

simply for ease of illustration, and actual devices made using these techniques will exhibit rounded corners, surface roughness, and other features.

(24) The description uses the phrases “in an embodiment” or “in embodiments,” which may each refer to one or more of the same or different embodiments. Furthermore, the terms “comprising,” “including,” “having,” and the like, as used with respect to embodiments of the present disclosure, are synonymous. As used herein, a “package” and an “IC package” are synonymous, as are a “die” and an “IC die.” The terms “top” and “bottom” may be used herein to explain various features of the drawings, but these terms are simply for ease of discussion, and do not imply a desired or required orientation. As used herein, the term “insulating” means “electrically insulating,” unless otherwise specified.

(25) When used to describe a range of dimensions, the phrase “between X and Y” represents a range that includes X and Y. For convenience, the phrase “FIG. 17” may be used to refer to the collection of drawings of FIGS. 17A-17F, the phrase “FIG. 18” may be used to refer to the collection of drawings of FIGS. 18A-18B, etc. Although certain elements may be referred to in the singular herein, such elements may include multiple sub-elements. For example, “an insulating material” may include one or more insulating materials. As used herein, a “conductive contact” may refer to a portion of conductive material (e.g., metal) serving as an electrical interface between different components; conductive contacts may be recessed in, flush with, or extending away from a surface of a component, and may take any suitable form (e.g., a conductive pad or socket, or portion of a conductive line or via).

(26) FIG. 1 is a side, cross-sectional view of a microelectronic assembly **100**, in accordance with various embodiments. A number of elements are illustrated in FIG. 1 as included in the microelectronic assembly **100**, but a number of these elements may not be present in a microelectronic assembly **100**. For example, in various embodiments, the heat spreader **131**, the thermal interface material **129**, the mold material **127**, the die **114-3**, the die **114-4**, the second-level interconnects **137**, and/or the circuit board **133** may not be included. Further, FIG. 1 illustrates a number of elements that are omitted from subsequent drawings for ease of illustration, but may be included in any of the microelectronic assemblies **100** disclosed herein. Examples of such elements include the heat spreader **131**, the thermal interface material **129**, the mold material **127**, the second-level interconnects **137**, and/or the circuit board **133**. Many of the elements of the microelectronic assembly **100** of FIG. 1 are included in other ones of the accompanying figures; the discussion of these elements is not repeated when discussing these figures, and any of these elements may take any of the forms disclosed herein. In some embodiments, individual ones of the microelectronic assemblies **100** disclosed herein may serve as a system-in-package (SiP) in which multiple dies **114** having different functionality are included. In such embodiments, the microelectronic assembly **100** may be referred to as an SiP.

(27) The microelectronic assembly **100** may include a package substrate **102** coupled to a die **114-1** by die-to-package substrate (DTPS) interconnects **150-1**. In particular, the top surface of the package substrate **102** may include a set of conductive contacts **146**, and the bottom surface of the die **114-1** may include a set of conductive contacts **122**; the conductive contacts **122** at the bottom surface of the die **114-1** may be electrically and mechanically coupled to the conductive contacts **146** at the top surface of the package substrate **102** by the DTPS interconnects **150-1**. In the embodiment of FIG. 1, the top surface of the package substrate **102** includes a recess **108** in which the die **114-1** is at least partially disposed; the conductive contacts **146** to which the die **114-1** is coupled are located at the bottom of the recess **108**. In other embodiments, the die **114-1** may not be disposed in a recess (e.g., as discussed below with reference to FIGS. 9-11). Any of the conductive contacts disclosed herein (e.g., the conductive contacts **122**, **124**, **146**, **140**, and/or **135**) may include bond pads, posts, or any other suitable conductive contact, for example.

(28) The package substrate **102** may include an insulating material (e.g., a dielectric material formed in multiple layers, as known in the art) and one or more conductive pathways through the

dielectric material (e.g., including conductive traces and/or conductive vias, as shown). In some embodiments, the insulating material of the package substrate **102** may be a dielectric material, such as an organic dielectric material, a fire retardant grade **4** material (FR-4), bismaleimide triazine (BT) resin, polyimide materials, glass reinforced epoxy matrix materials, or low-k and ultra low-k dielectric (e.g., carbon-doped dielectrics, fluorine-doped dielectrics, porous dielectrics, and organic polymeric dielectrics). In particular, when the package substrate **102** is formed using standard printed circuit board (PCB) processes, the package substrate **102** may include FR-4, and the conductive pathways in the package substrate **102** may be formed by patterned sheets of copper separated by build-up layers of the FR-4. The conductive pathways in the package substrate **102** may be bordered by liner materials, such as adhesion liners and/or barrier liners, as suitable.

(29) In some embodiments, one or more of the conductive pathways in the package substrate **102** may extend between a conductive contact **146** at the top surface of the package substrate **102** and a conductive contact **140** at the bottom surface of the package substrate **102**. In some embodiments, one or more of the conductive pathways in the package substrate **102** may extend between a conductive contact **146** at the bottom of the recess **108** and a conductive contact **140** at the bottom surface of the package substrate **102**. In some embodiments, one or more of the conductive pathways in the package substrate **102** may extend between different conductive contacts **146** at the top surface of the package substrate **102** (e.g., between a conductive contact **146** at the bottom of the recess **108** and a different conductive contact **146** at the top surface of the package substrate **102**). In some embodiments, one or more of the conductive pathways in the package substrate **102** may extend between different conductive contacts **140** at the bottom surface of the package substrate **102**.

(30) The dies **114** disclosed herein may include an insulating material (e.g., a dielectric material formed in multiple layers, as known in the art) and multiple conductive pathways formed through the insulating material. In some embodiments, the insulating material of a die **114** may include a dielectric material, such as silicon dioxide, silicon nitride, oxynitride, polyimide materials, glass reinforced epoxy matrix materials, or a low-k or ultra low-k dielectric (e.g., carbon-doped dielectrics, fluorine-doped dielectrics, porous dielectrics, organic polymeric dielectrics, photo-imageable dielectrics, and/or benzocyclobutene-based polymers). In some embodiments, the insulating material of a die **114** may include a semiconductor material, such as silicon, germanium, or a III-V material (e.g., gallium nitride), and one or more additional materials. For example, an insulating material may include silicon oxide or silicon nitride. The conductive pathways in a die **114** may include conductive traces and/or conductive vias, and may connect any of the conductive contacts in the die **114** in any suitable manner (e.g., connecting multiple conductive contacts on a same surface or on different surfaces of the die **114**). Example structures that may be included in the dies **114** disclosed herein are discussed below with reference to FIG. **29**. The conductive pathways in the dies **114** may be bordered by liner materials, such as adhesion liners and/or barrier liners, as suitable.

(31) In some embodiments, the die **114-1** may include conductive pathways to route power, ground, and/or signals to/from some of the other dies **114** included in the microelectronic assembly **100**. For example, the die **114-1** may include through-substrate vias (TSVs, including a conductive material via, such as a metal via, isolated from the surrounding silicon or other semiconductor material by a barrier oxide) or other conductive pathways through which power, ground, and/or signals may be transmitted between the package substrate **102** and one or more dies **114** “on top” of the die **114-1** (e.g., in the embodiment of FIG. **1**, the die **114-2** and/or the die **114-3**). In some embodiments, the die **114-1** may include conductive pathways to route power, ground, and/or signals between different ones of the dies **114** “on top” of the die **114-1** (e.g., in the embodiment of FIG. **1**, the die **114-2** and the die **114-3**). In some embodiments, the die **114-1** may be the source and/or destination of signals communicated between the die **114-1** and other dies **114** included in the microelectronic assembly **100**.

(32) In some embodiments, the die **114-1** may not route power and/or ground to the die **114-2**; instead, the die **114-2** may couple directly to power and/or ground lines in the package substrate **102**. By allowing the die **114-2** to couple directly to power and/or ground lines in the package substrate **102**, such power and/or ground lines need not be routed through the die **114-1**, allowing the die **114-1** to be made smaller or to include more active circuitry or signal pathways.

(33) In some embodiments, the die **114-1** may only include conductive pathways, and may not contain active or passive circuitry. In other embodiments, the die **114-1** may include active or passive circuitry (e.g., transistors, diodes, resistors, inductors, and capacitors, among others). In some embodiments, the die **114-1** may include one or more device layers including transistors (e.g., as discussed below with reference to FIG. 29. When the die **114-1** includes active circuitry, power and/or ground signals may be routed through the package substrate **102** and to the die **114-1** through the conductive contacts **122** on the bottom surface of the die **114-1**.

(34) Although FIG. 1 illustrates a specific number and arrangement of conductive pathways in the package of **102** and/or one or more of the dies **114**, these are simply illustrative, and any suitable number and arrangement may be used. The conductive pathways disclosed herein (e.g., conductive traces and/or conductive vias) may be formed of any appropriate conductive material, such as copper, silver, nickel, gold, aluminum, or other metals or alloys, for example.

(35) In some embodiments, the package substrate **102** may be a lower density medium and the die **114-1** may be a higher density medium. As used herein, the term “lower density” and “higher density” are relative terms indicating that the conductive pathways (e.g., including conductive lines and conductive vias) in a lower density medium are larger and/or have a greater pitch than the conductive pathways in a higher density medium. In some embodiments, a higher density medium may be manufactured using a modified semi-additive process or a semi-additive build-up process with advanced lithography (with small vertical interconnect features formed by advanced laser or lithography processes), while a lower density medium may be a PCB manufactured using a standard PCB process (e.g., a standard subtractive process using etch chemistry to remove areas of unwanted copper, and with coarse vertical interconnect features formed by a standard laser process).

(36) The microelectronic assembly **100** of FIG. 1 may also include a die **114-2**. The die **114-2** may be electrically and mechanically coupled to the package substrate **102** by DTPS interconnects **150-2**, and may be electrically and mechanically coupled to the die **114-1** by die-to-die (DTD) interconnects **130-1**. In particular, the top surface of the package substrate **102** may include a set of conductive contacts **146**, and the bottom surface of the die **114-2** may include a set of conductive contacts **122**; the conductive contacts **122** at the bottom surface of the die **114-1** may be electrically and mechanically coupled to the conductive contacts **146** at the top surface of the package substrate **102** by the DTPS interconnects **150-2**. Further, the top surface of the die **114-1** may include a set of conductive contacts **124**, and the bottom surface of the die **114-2** may include a set of conductive contacts **124**; the conductive contacts **124** at the bottom surface of the die **114-2** may be electrically and mechanically coupled to some of the conductive contacts **124** at the top surface of the die **114-1** by the DTD interconnects **130-1**. FIG. 2 is a bottom view of the die **114-2** of the microelectronic assembly **100** of FIG. 1, showing the “coarser” conductive contacts **122** and the “finer” conductive contacts **124**. The die **114-2** of the microelectronic assembly **100** may thus be a single-sided die (in the sense that the die **114-2** only has conductive contacts **122/124** on a single surface), and may be a mixed-pitch die (in the sense that the die **114-2** has sets of conductive contacts **122/124** with different pitch). Although FIG. 2 illustrates the conductive contacts **122** and the conductive contacts **124** as each being arranged in a rectangular array, this need not be the case, and the conductive contacts **122** and **124** may be arranged in any suitable pattern (e.g., triangular, hexagonal, rectangular, different arrangements between the conductive contacts **122** and **124**, etc.). A die **114** that has DTPS interconnects **150** and DTD interconnects **130** at the same surface may be referred to as a mixed-pitch die **114**; more generally, a die **114** that has interconnects **130** of



different pitches at a same surface may be referred to as a mixed-pitch die **114**.

(37) The die **114-2** may extend over the die **114-1** by an overlap distance **191**. In some embodiments, the overlap distance **191** may be between 0.5 millimeters and 5 millimeters (e.g., between 0.75 millimeters and 2 millimeters, or approximately 1 millimeter).

(38) The microelectronic assembly **100** of FIG. **1** may also include a die **114-3**. The die **114-3** may be electrically and mechanically coupled to the die **114-1** by DTD interconnects **130-2**. In particular, the bottom surface of the die **114-3** may include a set of conductive contacts **124** that are electrically and mechanically coupled to some of the conductive contacts **124** at the top surface of the die **114-1** by the DTD interconnects **130-2**. In the embodiment of FIG. **1**, the die **114-3** may be a single-sided, single-pitch die; in other embodiments, the die **114-3** may be a double-sided (or “multi-level,” or “omni-directional”) die, and additional components may be disposed on the top surface of the die **114-3**.

(39) As discussed above, in the embodiment of FIG. **1**, the die **114-1** may provide high density interconnect routing in a localized area of the microelectronic assembly **100**. In some embodiments, the presence of the die **114-1** may support direct chip attach of fine-pitch semiconductor dies (e.g., the dies **114-2** and **114-3**) that cannot be attached entirely directly to the package substrate **102**. In particular, as discussed above, the die **114-1** may support trace widths and spacings that are not achievable in the package substrate **102**. The proliferation of wearable and mobile electronics, as well as Internet of Things (IoT) applications, are driving reductions in the size of electronic systems, but limitations of the PCB manufacturing process and the mechanical consequences of thermal expansion during use have meant that chips having fine interconnect pitch cannot be directly mounted to a PCB. Various embodiments of the microelectronic assemblies **100** disclosed herein may be capable of supporting chips with high density interconnects and chips with low-density interconnects without sacrificing performance or manufacturability.

(40) The microelectronic assembly **100** of FIG. **1** may also include a die **114-4**. The die **114-4** may be electrically and mechanically coupled to the package substrate **102** by DTPS interconnects **150-3**. In particular, the bottom surface of the die **114-4** may include a set of conductive contacts **122** that are electrically and mechanically coupled to some of the conductive contacts **146** at the top surface of the package substrate **102** by the DTPS interconnects **150-3**. In the embodiment of FIG. **1**, the die **114-4** may be a single-sided, single-pitch die; in other embodiments, the die **114-4** may be a double-sided die, and additional components may be disposed on the top surface of the die **114-4**. Additional passive components, such as surface-mount resistors, capacitors, and/or inductors, may be disposed on the top surface or the bottom surface of the package substrate **102**, or embedded in the package substrate **102**.

(41) The microelectronic assembly **100** of FIG. **1** may also include a circuit board **133**. The package substrate **102** may be coupled to the circuit board **133** by second-level interconnects **137** at the bottom surface of the package substrate **102**. In particular, the package substrate **102** may include conductive contacts **140** at its bottom surface, and the circuit board **133** may include conductive contacts **135** at its top surface; the second-level interconnects **137** may electrically and mechanically couple the conductive contacts **135** and the conductive contacts **140**. The second-level interconnects **137** illustrated in FIG. **1** are solder balls (e.g., for a ball grid array arrangement), but any suitable second-level interconnects **137** may be used (e.g., pins in a pin grid array arrangement or lands in a land grid array arrangement). The circuit board **133** may be a motherboard, for example, and may have other components attached to it (not shown). The circuit board **133** may include conductive pathways and other conductive contacts (not shown) for routing power, ground, and signals through the circuit board **133**, as known in the art. In some embodiments, the second-level interconnects **137** may not couple the package substrate **102** to a circuit board **133**, but may instead couple the package substrate **102** to another IC package, an interposer, or any other suitable component.

(42) The microelectronic assembly **100** of FIG. **1** may also include a mold material **127**. The mold

material **127** may extend around one or more of the dies **114** on the package substrate **102**. In some embodiments, the mold material **127** may extend above one or more of the dies **114** on the package substrate **102**. In some embodiments, the mold material **127** may extend between one or more of the dies **114** and the package substrate **102** around the associated DTPS interconnects **150**; in such embodiments, the mold material **127** may serve as an underfill material. In some embodiments, the mold material **127** may extend between different ones of the dies **114** around the associated DTD interconnects **130**; in such embodiments, the mold material **127** may serve as an underfill material. The mold material **127** may include multiple different mold materials (e.g., an underfill material, and a different overmold material). The mold material **127** may be an insulating material, such as an appropriate epoxy material. In some embodiments, the mold material **127** may include an underfill material that is an epoxy flux that assists with soldering the dies **114-1/114-2** to the package substrate **102** when forming the DTPS interconnects **150-1** and **150-2**, and then polymerizes and encapsulates the DTPS interconnects **150-1** and **150-2**. The mold material **127** may be selected to have a coefficient of thermal expansion (CTE) that may mitigate or minimize the stress between the dies **114** and the package substrate **102** arising from uneven thermal expansion in the microelectronic assembly **100**. In some embodiments, the CTE of the mold material **127** may have a value that is intermediate to the CTE of the package substrate **102** (e.g., the CTE of the dielectric material of the package substrate **102**) and a CTE of the dies **114**.

(43) The microelectronic assembly **100** of FIG. **1** may also include a thermal interface material (TIM) **129**. The TIM **129** may include a thermally conductive material (e.g., metal particles) in a polymer or other binder. The TIM **129** may be a thermal interface material paste or a thermally conductive epoxy (which may be a fluid when applied and may harden upon curing, as known in the art). The TIM **129** may provide a path for heat generated by the dies **114** to readily flow to the heat spreader **131**, where it may be spread and/or dissipated. Some embodiments of the microelectronic assembly **100** of FIG. **1** may include a sputtered back side metallization (not shown) across the mold material **127** and the dies **114**; the TIM **129** (e.g., a solder TIM) may be disposed on this back side metallization.

(44) The microelectronic assembly **100** of FIG. **1** may also include a heat spreader **131**. The heat spreader **131** may be used to move heat away from the dies **114** (e.g., so that the heat may be more readily dissipated by a heat sink or other thermal management device). The heat spreader **131** may include any suitable thermally conductive material (e.g., metal, appropriate ceramics, etc.), and may include any suitable features (e.g., fins). In some embodiments, the heat spreader **131** may be an integrated heat spreader.

(45) The DTPS interconnects **150** disclosed herein may take any suitable form. In some embodiments, a set of DTPS interconnects **150** may include solder (e.g., solder bumps or balls that are subject to a thermal reflow to form the DTPS interconnects **150**). DTPS interconnects **150** that include solder may include any appropriate solder material, such as lead/tin, tin/bismuth, eutectic tin/silver, ternary tin/silver/copper, eutectic tin/copper, tin/nickel/copper, tin/bismuth/copper, tin/indium/copper, tin/zinc/indium/bismuth, or other alloys. In some embodiments, a set of DTPS interconnects **150** may include an anisotropic conductive material, such as an anisotropic conductive film or an anisotropic conductive paste. An anisotropic conductive material may include conductive materials dispersed in a non-conductive material. In some embodiments, an anisotropic conductive material may include microscopic conductive particles embedded in a binder or a thermoset adhesive film (e.g., a thermoset biphenyl-type epoxy resin, or an acrylic-based material). In some embodiments, the conductive particles may include a polymer and/or one or more metals (e.g., nickel or gold). For example, the conductive particles may include nickel-coated gold or silver-coated copper that is in turn coated with a polymer. In another example, the conductive particles may include nickel. When an anisotropic conductive material is uncompressed, there may be no conductive pathway from one side of the material to the other. However, when the anisotropic conductive material is adequately compressed (e.g., by conductive contacts on either

side of the anisotropic conductive material), the conductive materials near the region of compression may contact each other so as to form a conductive pathway from one side of the film to the other in the region of compression.

(46) The DTD interconnects **130** disclosed herein may take any suitable form. The DTD interconnects **130** may have a finer pitch than the DTPS interconnects **150** in a microelectronic assembly. In some embodiments, the dies **114** on either side of a set of DTD interconnects **130** may be unpackaged dies, and/or the DTD interconnects **130** may include small conductive bumps or pillars (e.g., copper bumps or pillars) attached to the conductive contacts **124** by solder. The DTD interconnects **130** may have too fine a pitch to couple to the package substrate **102** directly (e.g., too fine to serve as DTPS interconnects **150**). In some embodiments, a set of DTD interconnects **130** may include solder. DTD interconnects **130** that include solder may include any appropriate solder material, such as any of the materials discussed above. In some embodiments, a set of DTD interconnects **130** may include an anisotropic conductive material, such as any of the materials discussed above. In some embodiments, the DTD interconnects **130** may be used as data transfer lanes, while the DTPS interconnects **150** may be used for power and ground lines, among others.

(47) In some embodiments, some or all of the DTD interconnects **130** in a microelectronic assembly **100** may be metal-to-metal interconnects (e.g., copper-to-copper interconnects, or plated interconnects). In such embodiments, the conductive contacts **124** on either side of the DTD interconnect **130** may be bonded together (e.g., under elevated pressure and/or temperature) without the use of intervening solder or an anisotropic conductive material. In some embodiments, a thin cap of solder may be used in a metal-to-metal interconnect to accommodate planarity, and this solder may become an intermetallic compound during processing. In some metal-to-metal interconnects that utilize hybrid bonding, a dielectric material (e.g., silicon oxide, silicon nitride, silicon carbide, or an organic layer) may be present between the metals bonded together (e.g., between copper pads or posts that provide the associated conductive contacts **124**). In some embodiments, one side of a DTD interconnect **130** may include a metal pillar (e.g., a copper pillar), and the other side of the DTD interconnect may include a metal contact (e.g., a copper contact) recessed in a dielectric. In some embodiments, a metal-to-metal interconnect (e.g., a copper-to-copper interconnect) may include a noble metal (e.g., gold) or a metal whose oxides are conductive (e.g., silver). In some embodiments, a metal-to-metal interconnect may include metal nanostructures (e.g., nanorods) that may have a reduced melting point. Metal-to-metal interconnects may be capable of reliably conducting a higher current than other types of interconnects; for example, some solder interconnects may form brittle intermetallic compounds when current flows, and the maximum current provided through such interconnects may be constrained to mitigate mechanical failure.

(48) In some embodiments, some or all of the DTD interconnects **130** in a microelectronic assembly **100** may be solder interconnects that include a solder with a higher melting point than a solder included in some or all of the DTPS interconnects **150**. For example, when the DTD interconnects **130** in a microelectronic assembly **100** are formed before the DTPS interconnects **150** are formed (e.g., as discussed below with reference to FIGS. **17A-17F**), solder-based DTD interconnects **130** may use a higher-temperature solder (e.g., with a melting point above 200 degrees Celsius), while the DTPS interconnects **150** may use a lower-temperature solder (e.g., with a melting point below 200 degrees Celsius). In some embodiments, a higher-temperature solder may include tin; tin and gold; or tin, silver, and copper (e.g., 96.5% tin, 3% silver, and 0.5% copper). In some embodiments, a lower-temperature solder may include tin and bismuth (e.g., eutectic tin bismuth) or tin, silver, and bismuth. In some embodiments, a lower-temperature solder may include indium, indium and tin, or gallium.

(49) In the microelectronic assemblies **100** disclosed herein, some or all of the DTPS interconnects **150** may have a larger pitch than some or all of the DTD interconnects **130**. DTD interconnects **130** may have a smaller pitch than DTPS interconnects **150** due to the greater similarity of materials in

the different dies **114** on either side of a set of DTD interconnects **130** than between the die **114** and the package substrate **102** on either side of a set of DTPS interconnects **150**. In particular, the differences in the material composition of a die **114** and a package substrate **102** may result in differential expansion and contraction of the die **114** and the package substrate **102** due to heat generated during operation (as well as the heat applied during various manufacturing operations). To mitigate damage caused by this differential expansion and contraction (e.g., cracking, solder bridging, etc.), the DTPS interconnects **150** may be formed larger and farther apart than DTD interconnects **130**, which may experience less thermal stress due to the greater material similarity of the pair of dies **114** on either side of the DTD interconnects. In some embodiments, the DTPS interconnects **150** disclosed herein may have a pitch between 80 microns and 300 microns, while the DTD interconnects **130** disclosed herein may have a pitch between 7 microns and 100 microns.

(50) The elements of the microelectronic assembly **100** may have any suitable dimensions. Only a subset of the accompanying figures are labeled with reference numerals representing dimensions, but this is simply for clarity of illustration, and any of the microelectronic assemblies **100** disclosed herein may have components having the dimensions discussed herein. For example, in some embodiments, the thickness **164** of the package substrate **102** may be between 0.1 millimeters and 1.4 millimeters (e.g., between 0.1 millimeters and 0.35 millimeters, between 0.25 millimeters and 0.8 millimeters, or approximately 1 millimeter). In some embodiments, the recess **108** may have a depth **175** between 10 microns and 200 microns (e.g., between 10 microns and 30 microns, between 30 microns and 100 microns, between 60 microns and 80 microns, or approximately 75 microns). In some embodiments, the depth **175** may be equal to a certain number of layers of the dielectric material in the package substrate **102**. For example, the depth **175** may be approximately equal to between one and five layers of the dielectric material in the package substrate **102** (e.g., two or three layers of the dielectric material). In some embodiments, the depth **175** may be equal to the thickness of a solder resist material (not shown) on the top surface of the package substrate **102**.

(51) In some embodiments, the distance **179** between the bottom surface of the die **114-1** and the proximate top surface of the package substrate **102** (at the bottom of the recess **108**) may be less than the distance **177** between the bottom surface of the die **114-2** and the proximate top surface of the package substrate **102**. In some embodiments, the distance **179** may be approximately the same as the distance **177**. In some embodiments, the distance **177** between the bottom surface of the die **114-2** and the proximate top surface of the package substrate **102** may be greater than the distance **193** between the bottom surface of the die **114-2** and the proximate top surface of the die **114-1**. In other embodiments, the distance **177** may be less than or equal to the distance **193**.

(52) In some embodiments, the top surface of the die **114-1** may extend higher than the top surface of the package substrate **102**, as illustrated in FIG. 1. In other embodiments, the top surface of the die **114-1** may be substantially coplanar with the top surface of the package substrate **102**, or may be recessed below the top surface of the package substrate **102**. FIG. 3 illustrates an example of the former embodiment. Although various ones of the figures illustrate microelectronic assemblies **100** having a single recess **108** in the package substrate **102**, the thickness of **102** may include multiple recesses **108** (e.g., having the same or different dimensions, and each having a die **114** disposed therein), or no recesses **108**. Examples of the former embodiments are discussed below with reference to FIGS. 7-8, and examples of the latter embodiments are discussed below with reference to FIGS. 9-11. In some embodiments, a recess **108** may be located at the bottom surface of the package substrate **102** (e.g., proximate to the conductive contacts **140**), instead of or in addition to a recess **108** at the top surface of the package substrate **102**.

(53) In the embodiment of FIG. 1, a single die **114-2** is illustrated as “spanning” the package substrate **102** and the die **114-1**. In some embodiments of the microelectronic assemblies **100** disclosed herein, multiple dies **114** may span the package substrate **102** and another die **114**. For example, FIG. 4 illustrates an embodiment in which two dies **114-2** each have conductive contacts **122** and conductive contacts **124** disposed at the bottom surfaces; the conductive contacts **122** of

the dies **114-2** are coupled to conductive contacts **146** at the top surface of the package substrate **102** via DTPS interconnects **150-2**, and the conductive contacts **124** of the dies **114-2** are coupled to conductive contacts **124** at the top surface of the die **114** via DTD interconnects **130**. In some embodiments, power and/or ground signals may be provided directly to the dies **114** of the microelectronic assembly **100** of FIG. 4 through the package substrate **102**, and the die **114-1** may, among other things, route signals between the dies **114-2**.

(54) In some embodiments, the die **114-1** may be arranged as a bridge between multiple other dies **114**, and may also have additional dies **114** disposed thereon. For example, FIG. 5 illustrates an embodiment in which two dies **114-2** each have conductive contacts **122** and conductive contacts **124** disposed at the bottom surfaces; the conductive contacts **122** of the dies **114-2** are coupled to conductive contacts **146** at the top surface of the package substrate **102** via DTPS interconnects **150-2**, and the conductive contacts **124** of the dies **114-2** are coupled to conductive contacts **124** at the top surface of the die **114** via DTD interconnects **130** (e.g., as discussed above with reference to FIG. 4). Additionally, a die **114-3** (or multiple dies **114-3**, not shown) is coupled to the die **114-1** by conductive contacts **124** on proximate surfaces of these dies **114** and intervening DTD interconnects **130-2** (e.g., as discussed above with reference to FIG. 1).

(55) As noted above, any suitable number of the dies **114** in a microelectronic assembly **100** may be double-sided dies **114**. For example, FIG. 6 illustrates a microelectronic assembly **100** sharing a number of elements with FIG. 1, but including a double-sided die **114-6**. The die **114-6** includes conductive contacts **122** and **124** at its bottom surface; the conductive contacts **122** at the bottom surface of the die **114-6** are coupled to conductive contacts **146** at the top surface of the package substrate **102** via DTPS interconnects **150-2**, and the conductive contacts **124** at the bottom surface of the die **114-6** are coupled to conductive contacts **124** at the top surface of the die **114-1** via DTD interconnects **130-1**. The die **114-6** also includes conductive contacts **124** at its top surface; these conductive contacts **124** are coupled to conductive contacts **124** at the bottom surface of a die **114-7** by DTD interconnects **130-3**.

(56) As noted above, a package substrate **102** may include one or more recesses **108** in which dies **114** are at least partially disposed. For example, FIG. 7 illustrates a microelectronic assembly **100** including a package substrate **102** having two recesses: a recess **108-1** and a recess **108-2**. In the embodiment of FIG. 7, the recess **108-1** is nested in the recess **108-2**, but in other embodiments, multiple recesses **108** need not be nested. In FIG. 7, the die **114-1** is at least partially disposed in the recess **108-1**, and the dies **114-6** and **114-3** are at least partially disposed in the recess **108-2**. In the embodiment of FIG. 7, like the embodiment of FIG. 6, the die **114-6** includes conductive contacts **122** and **124** at its bottom surface; the conductive contacts **122** at the bottom surface of the die **114-6** are coupled to conductive contacts **146** at the top surface of the package substrate **102** via DTPS interconnects **150-2**, and the conductive contacts **124** at the bottom surface of the die **114-6** are coupled to conductive contacts **124** at the top surface of the die **114-1** via DTD interconnects **130-1**. The die **114-6** also includes conductive contacts **124** at its top surface; these conductive contacts **124** are coupled to conductive contacts **124** at the bottom surface of a die **114-7** by DTD interconnects **130-3**. Further, the microelectronic assembly **100** of FIG. 7 includes a die **114-8** that spans the package substrate **102** and the die **114-6**. In particular, the die **114-8** includes conductive contacts **122** and **124** at its bottom surface; the conductive contacts **122** at the bottom surface of the die **114-8** are coupled to conductive contacts **146** at the top surface of the package substrate **102** via DTPS interconnects **150-3**, and the conductive contacts **124** at the bottom surface of the die **114-8** are coupled to conductive contacts **124** at the top surface of the die **114-6** via DTD interconnects **130-4**.

(57) In various ones of the microelectronic assemblies **100** disclosed herein, a single die **114** may bridge to other dies **114** from “below” (e.g., as discussed above with reference to FIGS. 4 and 5) or from “above.” For example, FIG. 8 illustrates a microelectronic assembly **100** similar to the microelectronic assembly **100** of FIG. 7, but including two double-sided dies **114-9** and **114-10**, as

well as an additional die **114-11**. The die **114-9** includes conductive contacts **122** and **124** at its bottom surface; the conductive contacts **122** at the bottom surface of the die **114-9** are coupled to conductive contacts **146** at the top surface of the package substrate **102** via DTPS interconnects **150-3**, and the conductive contacts **124** at the bottom surface of the die **114-9** are coupled to conductive contacts **124** at the top surface of the die **114-6** via DTD interconnects **130-4**. The die **114-6** includes conductive contacts **124** at its top surface; these conductive contacts **124** are coupled to conductive contacts **124** at the bottom surface of a die **114-10** by DTD interconnects **130-3**. Further, the die **114-11** includes conductive contacts **124** at its bottom surface; some of these conductive contacts **124** are coupled to conductive contacts **124** at the top surface of the die **114-9** by DTD interconnects **130-6**, and some of these conductive contacts **124** are coupled to conductive contacts **124** at the top surface of the die **114-10** by DTD interconnects **130-5**. The die **114-11** may thus bridge the dies **114-9** and **114-10**.

(58) As noted above, in some embodiments, the package substrate **102** may not include any recesses **108**. For example, FIG. **9** illustrates an embodiment having dies **114** and a package substrate **102** mutually interconnected in the manner discussed above with reference to FIG. **1**, but in which the die **114-1** is not disposed in a recess in the package substrate **102**. Instead, the dies **114** are disposed above a planar portion of the top surface of the package substrate **102**. Any suitable ones of the embodiments disclosed herein that include recesses **108** may have counterpart embodiments that do not include a recess **108**. For example, FIG. **10** illustrates a microelectronic assembly **100** having dies **114** and a package substrate **102** mutually interconnected in the manner discussed above with reference to FIG. **4**, but in which the die **114-1** is not disposed in a recess in the package substrate **102**.

(59) Any of the arrangements of dies **114** illustrated in any of the accompanying figures may be part of a repeating pattern in a microelectronic assembly **100**. For example, FIG. **11** illustrates a portion of a microelectronic assembly **100** in which an arrangement like the one of FIG. **10** is repeated, with multiple dies **114-1** and multiple dies **114-2**. The dies **114-1** may bridge the adjacent dies **114-2**. More generally, the microelectronic assemblies **100** disclosed herein may include any suitable arrangement of dies **114**. FIGS. **12-16** are top views of example arrangements of multiple dies **114** in various microelectronic assemblies **100**, in accordance with various embodiments. The package substrate **102** is omitted from FIGS. **12-16**; some or all of the dies **114** in these arrangements may be at least partially disposed in a recess **108** in a package substrate **102**, or may not be disposed in a recess of a package substrate **102**. In the arrangements of FIGS. **12-16**, the different dies **114** may include any suitable circuitry. For example, in some embodiments, the die **114A** may be an active or passive die, and the dies **114B** may include input/output circuitry, high bandwidth memory, and/or enhanced dynamic random access memory (EDRAM).

(60) FIG. **12** illustrates an arrangement in which a die **114A** is disposed below multiple different dies **114B**. The die **114A** may be connected to a package substrate **102** (not shown) in any of the manners disclosed herein with reference to the die **114-1**, while the dies **114B** may span the package substrate **102** and the die **114A** (e.g., in any of the manners disclosed herein with reference to the die **114-2**). FIG. **12** also illustrates a die **114C** disposed on the die **114A** (e.g., in the manner disclosed herein with reference to the die **114-3**). In FIG. **12**, the dies **114B** “overlap” the edges and/or the corners of the die **114A**, while the die **114C** is wholly above the die **114A**. Placing dies **114B** at least partially over the corners of the die **114A** may reduce routing congestion in the die **114A** and may improve utilization of the die **114A** (e.g., in case the number of input/outputs needed between the die **114A** and the dies **114B** is not large enough to require the full edge of the die **114A**). In some embodiments, the die **114A** may be disposed in a recess **108** in a package substrate **102**. In some embodiments, the die **114A** may be disposed in a recess **108** in a package substrate **102**, and the dies **114B** may be disposed in one or more recesses **108** in the package substrate **102**. In some embodiments, none of the dies **114A** or **114B** may be disposed in recesses **108**.

(61) FIG. **13** illustrates an arrangement in which a die **114A** is disposed below multiple different

dies **114B**. The die **114A** may be connected to a package substrate **102** (not shown) in any of the manners disclosed herein with reference to the die **114-1**, while the dies **114B** may span the package substrate **102** and the die **114A** (e.g., in any of the manners disclosed herein with reference to the die **114-2**). FIG. **13** also illustrates dies **114C** disposed on the die **114A** (e.g., in the manner disclosed herein with reference to the die **114-3**). In FIG. **13**, the dies **114B** “overlap” the edges of the die **114A**, while the dies **114C** are wholly above the die **114A**. In some embodiments, the die **114A** may be disposed in a recess **108** in a package substrate **102**. In some embodiments, the die **114A** may be disposed in a recess **108** in a package substrate **102**, and the dies **114B** may be disposed in one or more recesses **108** in the package substrate **102**. In some embodiments, none of the dies **114A** or **114B** may be disposed in recesses **108**. In the embodiment of FIG. **13**, the dies **114B** and **114C** may be arranged in a portion of a rectangular array. In some embodiments, two dies **114A** may take the place of the single die **114A** illustrated in FIG. **13**, and one or more dies **114C** may “bridge” the two dies **114A** (e.g., in the manner discussed below with reference to FIG. **15**).

(62) FIG. **14** illustrates an arrangement in which a die **114A** is disposed below multiple different dies **114B**. The die **114A** may be connected to a package substrate **102** (not shown) in any of the manners disclosed herein with reference to the die **114-1**, while the dies **114B** may span the package substrate **102** and the die **114A** (e.g., in any of the manners disclosed herein with reference to the die **114-2**). In FIG. **14**, the dies **114B** “overlap” the edges and/or the corners of the die **114A**. In some embodiments, the die **114A** may be disposed in a recess **108** in a package substrate **102**. In some embodiments, the die **114A** may be disposed in a recess **108** in a package substrate **102**, and the dies **114B** may be disposed in one or more recesses **108** in the package substrate **102**. In some embodiments, none of the dies **114A** or **114B** may be disposed in recesses **108**. In the embodiment of FIG. **14**, the dies **114B** may be arranged in a portion of a rectangular array.

(63) FIG. **15** illustrates an arrangement in which multiple dies **114A** are disposed below multiple different dies **114B** such that each die **114A** bridges two or more horizontally or vertically adjacent dies **114B**. The dies **114A** may be connected to a package substrate **102** (not shown) in any of the manners disclosed herein with reference to the die **114-1**, while the dies **114B** may span the package substrate **102** and the die **114A** (e.g., in any of the manners disclosed herein with reference to the die **114-2**). In FIG. **12**, the dies **114B** “overlap” the edges of the adjacent dies **114A**. In some embodiments, the dies **114A** may be disposed in one or more recesses **108** in a package substrate **102**. In some embodiments, the dies **114A** may be disposed in one or more recesses **108** in a package substrate **102**, and the dies **114B** may be disposed in one or more recesses **108** in the package substrate **102**. In some embodiments, none of the dies **114A** or **114B** may be disposed in recesses **108**. In FIG. **15**, the dies **114A** and the dies **114B** may be arranged in rectangular arrays.

(64) FIG. **16** illustrates an arrangement in which multiple dies **114A** are disposed below multiple different dies **114B** such that each die **114A** bridges the four diagonally adjacent dies **114B**. The dies **114A** may be connected to a package substrate **102** (not shown) in any of the manners disclosed herein with reference to the die **114-1**, while the dies **114B** may span the package substrate **102** and the die **114A** (e.g., in any of the manners disclosed herein with reference to the die **114-2**). In FIG. **12**, the dies **114B** “overlap” the corners of the adjacent dies **114A**. In some embodiments, the dies **114A** may be disposed in one or more recesses **108** in a package substrate **102**. In some embodiments, the dies **114A** may be disposed in one or more recesses **108** in a package substrate **102**, and the dies **114B** may be disposed in one or more recesses **108** in the package substrate **102**. In some embodiments, none of the dies **114A** or **114B** may be disposed in recesses **108**. In FIG. **16**, the dies **114A** and the dies **114B** may be arranged in rectangular arrays.

(65) Any suitable techniques may be used to manufacture the microelectronic assemblies disclosed herein. For example, FIGS. **17A-17F** are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly **100** of FIG. **5**, in accordance with various embodiments. Although the operations discussed below with reference to FIGS. **17A-17F** (and others of the accompanying drawings representing manufacturing processes) are illustrated in

a particular order, these operations may be performed in any suitable order. Additionally, although particular assemblies are illustrated in FIGS. 17A-17F (and others of the accompanying drawings representing manufacturing processes), the operations discussed below with reference to FIGS. 17A-17F may be used to form any suitable assemblies. In some embodiments, microelectronic assemblies 100 manufactured in accordance with the process of FIGS. 17A-17F (e.g., any of the microelectronic assemblies 100 of FIGS. 1-11) may have DTPS interconnects 150-1 that are solder interconnects, and DTD interconnects 130-1 and 130-2 that are non-solder interconnects (e.g., metal-to-metal interconnects or anisotropic conductive material interconnects). In the embodiment of FIGS. 17A-17F, the dies 114 may first be assembled into a “composite die,” and then the composite die may be coupled to the package substrate 102. This approach may allow for tighter tolerances in the formation of the DTD interconnects 130, and may be particularly desirable for relatively small dies 114.

(66) FIG. 17A illustrates an assembly 300 including a carrier 202 on which the dies 114-2 and 114-3 are disposed. The dies 114-2 and 114-3 are “upside down” on the carrier 202, in the sense that the conductive contacts 122 and 124 of the dies 114 are facing away from the carrier 202, and the conductive contacts 124 of the die 114-3 are facing away from the carrier 202. The dies 114-2 and 114-3 may be secured to the carrier using any suitable technique, such as a removable adhesive. The carrier 202 may include any suitable material for providing mechanical stability during subsequent manufacturing operations.

(67) FIG. 17B illustrates an assembly 302 subsequent to coupling the die 114-1 to the dies 114-2 and 114-3. In particular, the die 114-1 may be arranged “upside down” in the assembly 302 such that the conductive contacts 124 of the die 114-1 may be coupled to the conductive contacts 124 of the dies 114-2 (via DTD interconnects 130-1) and to the conductive contacts 124 of the die 114-3 (via DTD interconnects 130-2). Any suitable technique may be used to form the DTD interconnects 130 of the assembly 302, such as metal-to-metal attachment techniques, solder techniques, or anisotropic conductive material techniques.

(68) FIG. 17C illustrates an assembly 304 including a package substrate 203. The package substrate 203 may be structurally similar to the package substrate 102 of FIG. 5, but may not include the recess 108 of the package substrate 102. In some embodiments, the package substrate 203 may be manufactured using standard PCB manufacturing processes, and thus the package substrate 203 may take the form of a PCB, as discussed above. In some embodiments, the package substrate 203 may be a set of redistribution layers formed on a panel carrier (not shown) by laminating or spinning on a dielectric material, and creating conductive vias and lines by laser drilling and plating. Any method known in the art for fabrication of the package substrate 203 may be used, and for the sake of brevity, such methods will not be discussed in further detail herein.

(69) FIG. 17D illustrates an assembly 306 subsequent to forming a recess 108 in the package substrate 203 (FIG. 17C) to form the package substrate 102. The recess 108 may have a bottom surface at which conductive contacts 146 are exposed. Any suitable technique may be used to form the recess 108. For example, in some embodiments, the recess 108 may be laser-drilled down to a planar metal stop in the package substrate 203 (not shown); once the metal stop is reached, the metal stop may be removed to expose the conductive contacts 146 at the bottom of the recess 108. In some embodiments, the recess 108 may be formed by a mechanical drill.

(70) FIG. 17E illustrates an assembly 308 subsequent to “flipping” the assembly 302 (FIG. 17B) and bringing the dies 114-1 and 114-2 into alignment with the package substrate 102 (FIG. 17D) so that the conductive contacts 122 on the dies 114-1 and 114-2 are aligned with their respective conductive contacts 146 on the top surface of the package substrate 102.

(71) FIG. 17F illustrates an assembly 310 subsequent to forming DTPS interconnects 150 between the dies 114-1/114-2 and the package substrate 102 of the assembly 308 (FIG. 17E), then removing the carrier. The DTPS interconnects 150 may take any of the forms disclosed herein (e.g., solder interconnects, or anisotropic conductive material interconnects), and any suitable techniques may



be used to form the DTPS interconnects **150** (e.g., a mass reflow process or a thermal compression bonding process). The assembly **310** may take the form of the microelectronic assembly **100** of FIG. 5. Further operations may be performed as suitable (e.g., providing a mold material **127**, providing a TIM **129**, providing a heat spreader **131**, attaching additional dies **114** to the package substrate **102**, etc.).

(72) FIGS. **18A-18B** are side, cross-sectional views of various stages in another example process for manufacturing the microelectronic assembly **100** of FIG. 5, in accordance with various embodiments. In some embodiments, microelectronic assemblies **100** manufactured in accordance with the process of FIGS. **18A-18B** (e.g., any of the microelectronic assemblies **100** of FIGS. **1-11**) may have DTPS interconnects **150-1** that are solder interconnects, and DTD interconnects **130-1** and **130-2** that are also solder interconnects. In the embodiment of FIGS. **18A-18B**, the die **114-1** may be coupled to the package substrate **102**, and then the remaining dies **114** may be attached. This approach may accommodate the tolerance and warpage of the package substrate **102**, and may be particularly desirable for relatively larger dies **114**. The process of FIGS. **17A-17F** may advantageously be more compatible with non-solder DTD interconnects **130**, while the process of FIGS. **18A-18B** may advantageously involve simpler handling of the dies **114**.

(73) FIG. **18A** illustrates an assembly **312** subsequent to coupling the die **114-1** to the package substrate **102**. In particular, the die **114-1** may be positioned in the recess **108**, and conductive contacts **122** at the bottom surface of the die **114-1** may be coupled to conductive contacts **146** at the top surface of the package substrate **102** by DTPS interconnects **150-1**. The DTPS interconnects **150-1** may take the form of any of the embodiments disclosed herein, such as solder interconnects or anisotropic conductive material interconnects. The package substrate **102** may be formed in accordance with any of the techniques discussed above with reference to FIGS. **17C-17D**.

(74) FIG. **18B** illustrates an assembly **314** subsequent to coupling the dies **114-2** and **114-3** to the assembly **312** (FIG. **18A**). In particular, the conductive contacts **124** of the die **114-1** may be coupled to the conductive contacts **124** of the dies **114-2** (via DTD interconnects **130-1**) and to the conductive contacts **124** of the die **114-3** (via DTD interconnects **130-2**). Further, the conductive contacts **122** of the dies **114-2** may be coupled to conductive contacts **146** at the top surface of the package substrate **102** via DTPS interconnects **150-2**. Any suitable technique may be used to form the DTD interconnects **130-1** and **130-2**, and the DTPS interconnects **150-2**, of the assembly **314**, such as solder techniques or anisotropic conductive material techniques. For example, the DTPS interconnects **150-2** and the DTD interconnects **130-1/130-2** may be solder interconnects. The assembly **314** may take the form of the microelectronic assembly **100** of FIG. 5. Further operations may be performed as suitable (e.g., providing a mold material **127**, providing a TIM **129**, providing a heat spreader **131**, attaching additional dies **114** to the package substrate **102**, etc.).

(75) FIGS. **19A-19H** are side, cross-sectional views of various stages in another example process for manufacturing the microelectronic assembly **100** of FIG. 5, in accordance with various embodiments. In some embodiments, microelectronic assemblies **100** manufactured in accordance with the process of FIGS. **19A-19H** (e.g., any of the microelectronic assemblies **100** of FIGS. **1-11**) may have DTPS interconnects **150-1** that are non-solder interconnects (e.g., anisotropic conductive material interconnects), and DTD interconnects **130-1** and **130-2** that are solder interconnects.

(76) FIG. **19A** illustrates an assembly **315** including a package substrate portion **113** on a carrier **202**. The package substrate portion **113** may be the “top” portion of the package substrate **102**, as discussed further below, and may include conductive contacts **146** at the surface of the package substrate portion **113** facing away from the carrier **202**. The carrier **202** may take any of the forms disclosed herein. The package substrate portion **113** may be formed on the carrier **202** using any suitable technique, such as a redistribution layer technique.

(77) FIG. **19B** illustrates an assembly **316** subsequent to forming a cavity **111** in the package substrate portion **113** of the assembly **315** (FIG. **19A**). The cavity **111** may be formed using any of

the techniques discussed above with reference to the recess **108** of FIG. **17D**, for example. As discussed in further detail below, the cavity **111** may correspond to the recess **108**.

(78) FIG. **19C** illustrates an assembly **318** subsequent to positioning the die **114-1** in the cavity **111** of the assembly **316** (FIG. **19B**). The die **114-1** may be positioned in the cavity **111** so that the conductive contacts **122** face the carrier **202**, and the conductive contacts **124** face away from the carrier **202**. In some embodiments, a pick-and-place machine may be used to position the die **114-1** in the cavity **111** on the carrier **202**.

(79) FIG. **19D** illustrates an assembly **320** subsequent to coupling the dies **114-2** and **114-3** to the assembly **318** (FIG. **19C**), and providing a mold material **127** around the dies **114**. In particular, the conductive contacts **124** of the die **114-1** may be coupled to the conductive contacts **124** of the dies **114-2** (via DTD interconnects **130-1**) and to the conductive contacts **124** of the die **114-3** (via DTD interconnects **130-2**). Further, the conductive contacts **122** of the dies **114-2** may be coupled to conductive contacts **146** at the top surface of the package substrate **102** via DTPS interconnects **150-2**. Any suitable technique may be used to form the DTD interconnects **130-1** and **130-2**, and the DTPS interconnects **150-2**, of the assembly **314**, such as solder techniques or anisotropic conductive material techniques. For example, the DTPS interconnects **150-2** and the DTD interconnects **130-1/130-2** may be solder interconnects. The mold material **127** may take any of the forms disclosed herein, and may provide mechanical support for further manufacturing operations.

(80) FIG. **19E** illustrates an assembly **321** subsequent to attaching another carrier **204** to the top surface of the assembly **320** (FIG. **19D**). The carrier **204** may take the form of any of the embodiments of the carrier **202** disclosed herein.

(81) FIG. **19F** illustrates an assembly **322** subsequent to removing the carrier **202** from the assembly **321** (FIG. **19E**) and flipping the result so that the package substrate portion **113** and the conductive contacts **122** of the die **114-1** are exposed.

(82) FIG. **19G** illustrates an assembly **324** subsequent to forming an additional package substrate portion **115** on the package substrate portion **113** of the assembly **322** (FIG. **19F**) to form the package substrate **102**. Any suitable technique may be used to form the package substrate portion **113**, including any of the techniques discussed above with reference to FIG. **19A**, a bumpless build-up layer technique, a carrier-based panel-level coreless package substrate manufacturing technique, or an embedded panel-level bonding technique. In some embodiments, forming the package substrate portion **115** may include plating the conductive contacts **122** of the die **114-1** with a metal or other conductive material as part of forming the proximate conductive contacts **146** of the package substrate **102**; consequently, the DTPS interconnects **150-1** between the die **114-1** and the package substrate **102** may be plated interconnects.

(83) FIG. **19H** illustrates an assembly **325** subsequent to removing the carrier **204** from the assembly **324** (FIG. **19G**) and flipping the result. The assembly **325** may take the form of the microelectronic assembly **100** of FIG. **5**. Further operations may be performed as suitable (e.g., providing a TIM **129**, providing a heat spreader **131**, attaching additional dies **114** to the package substrate **102**, etc.).

(84) In the microelectronic assemblies **100** discussed above with reference to FIGS. **1-11**, the die **114-1** is coupled directly to at least one die **114-2** without any intervening portion of the package substrate **102**. In other embodiments of the microelectronic assemblies **100** disclosed herein, a portion of the package substrate **102** may be disposed between an embedded die **114-1** and a die **114-2**. FIGS. **20-22** are side, cross-sectional views of example microelectronic assemblies **100** including such a feature, in accordance with various embodiments. In particular, FIGS. **20-22** illustrate arrangements of dies **114-1**, **114-2**, **114-3**, and **114-4** that are similar to the arrangement illustrated in FIG. **1**, but that further include a package substrate portion **148** between the top surface of the die **114-1** and the top surface of the package substrate **102**. The dies **114-2**, **114-3**, and **114-4** may all be coupled to this package substrate portion **148**. For example, the die **114-1** may include conductive contacts **122** at its bottom surface that couple to conductive contacts **146** of

the package substrate **102** via DTPS interconnects **150-1**, and the die **114-1** may include conductive contacts **122** at its top surface that couple to conductive contacts **146** of the package substrate **102** (in the package substrate portion **148**) via DTPS interconnects **150-4**.

(85) In some embodiments, the package substrate portion **148** may include one or more areas **149** with higher conductive pathway density (e.g., the areas in which the footprint of the die **114-2** overlaps with the footprint of the die **114-1** and the package substrate portion **148** includes conductive pathways between the die **114-2** and the die **114-1**, or the areas in which the footprint of the die **114-3** overlaps of the footprint of the die **114-1** and the package substrate portion **148** includes conductive pathways between the die **114-3** and the die **114-1**). Thus, the die **114-2** may be a mixed-pitch die including larger-pitch conductive contacts **122A** and smaller-pitch conductive contacts **122B**; the larger-pitch conductive contacts **122A** may couple (through some of the DTPS interconnects **150-2**) to conductive contacts **146** on the top surface of the package substrate **102** (that themselves couple to conductive pathways through the bulk of the package substrate **102**), and the smaller-pitch conductive contacts **122B** may couple (through some of the DTPS interconnects **150-2**) to conductive contacts **146** on the top surface of the package substrate **102** (that themselves couple to conductive pathways through the package substrate portion **148** and to the die **114-1**). Similarly, the pitch of the conductive contacts **122** at the bottom surface of the die **114-3** (which may be coupled via the DTPS interconnects **150-5** to dense conductive pathways through the package substrate portion **148** to the die **114-1**) may be smaller than the pitch of the conductive contacts **122** at the bottom surface of the die **114-4** (which may be coupled via the DTPS interconnects **150-3** to less dense conductive pathways through the package substrate **102**). The package substrate **102** may also include a portion **151** adjacent to the die **114-1**, and a portion **153** below the die **114-1**.

(86) FIG. **20** illustrates an embodiment in which the conductive pathways in the package substrate **102** are provided by conductive lines and vias, as known in the art. In other embodiments, the package substrate **102** may include conductive pillars (e.g., copper pillars) and other structures. For example, FIG. **21** illustrates a microelectronic assembly **100** similar to that of FIG. **20**, but in which the package substrate portion **151** includes a plurality of conductive pillars **134** disposed around the die **114-1**. The conductive pillars **134** may be substantially surrounded by a mold material **132**, which may take the form of any of the mold materials **127** disclosed herein. The conductive pillars **134** may be part of conductive pathways between the package substrate portion **148** and the package substrate portion **153**. Non-conductive pillars (e.g., pillars formed of a permanent resist or a dielectric) may be used instead of or in addition to conductive pillars **134** in any suitable ones of the embodiments disclosed herein.

(87) The conductive pillars **134** may be formed of any suitable conductive material, such as a metal. In some embodiments, the conductive pillars **134** may include copper. The conductive pillars **134** may have any suitable dimensions. For example, in some embodiments, an individual conductive pillar **134** may have an aspect ratio (height:diameter) between 1:1 and 4:1 (e.g., between 1:1 and 3:1). In some embodiments, an individual conductive pillar **134** may have a diameter between 10 microns and 300 microns. In some embodiments, an individual conductive pillar **134** may have a diameter between 50 microns and 400 microns.

(88) In some embodiments in which a package substrate **102** includes a plurality of conductive pillars **134**, the package substrate portion **151** may also include a placement ring. For example, FIG. **22** illustrates an embodiment of the microelectronic assembly **100** similar to that of FIG. **21**, but further including a placement ring **136**. The placement ring **136** may be formed of any suitable material (e.g., a plated copper feature with a coating of an organic material, stainless steel, or a non-conductive material, such as glass, sapphire, polyimide, or epoxy with silica), and may be shaped so as to fit closely around the die **114-1**. In some embodiments, the placement ring **136** may have slanted or straight walls to help guide the die **114-1** into position. Thus, the shape of the placement ring **136** may complement the shape of the footprint of the die **114-1**, and the placement

ring **136** may help to align the die **114-1** during manufacture, as discussed further below.

(89) Microelectronic assemblies **100** including embedded dies **114** may include any suitable arrangement of dies **114**. For example, any of the arrangements illustrated in FIGS. **12-16** may be implemented with the die **114A** embedded in a package substrate, with the dies **114A** and **114B** embedded in a package substrate **102**, or with the dies **114A**, **114B**, and **114C** embedded in a package substrate **102**. Additionally, any of the arrangements illustrated in FIGS. **1-11** may be implemented with the die **114-1** (and optionally more of the dies **114**) embedded in a package substrate **102**, in accordance with any of the embodiments of FIGS. **20-22**.

(90) Any suitable techniques may be used to manufacture microelectronic assemblies **100** having an embedded die **114-1** (e.g., having a package substrate portion **148** between the die **114-1** and the die **114-2**). For example, FIGS. **23A-23B** are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly **100** of FIG. **20**, in accordance with various embodiments. In some embodiments, microelectronic assemblies **100** manufactured in accordance with the process of FIGS. **23A-23B** may have DTSP interconnects **150-1** that are solder interconnects, and DTSP interconnects **150-4** that are non-solder interconnects (e.g., plated interconnects).

(91) FIG. **23A** illustrates an assembly **326** subsequent to forming the package substrate portion **148** on the assembly **312** (FIG. **18A**). The package substrate portion **148** may be formed using any suitable techniques, such as any of the techniques discussed above with reference to the formation of the package substrate portion **115** of FIG. **19G**. In some embodiments, forming the package substrate portion **148** may include plating the conductive contacts **122** of the die **114-1** with a metal or other conductive material as part of forming the proximate conductive contacts **146** of the package substrate **102**; consequently, the DTSP interconnects **150-4** between the die **114-1** and the package substrate portion **148** may be plated interconnects.

(92) FIG. **23B** illustrates an assembly **328** subsequent to attaching the dies **114-2**, **114-3**, and **114-4** to the assembly **326** (FIG. **23A**). Any suitable techniques may be used to form the DTSP interconnects **150** between the dies **114-2**, **114-3**, and **114-4** and the package substrate **102**, such as solder techniques or anisotropic conductive material techniques.

(93) FIGS. **24A-24E** are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly **100** of FIG. **21**, in accordance with various embodiments. In some embodiments, microelectronic assemblies **100** manufactured in accordance with the process of FIGS. **24A-24E** may have DTSP interconnects **150-1** that are solder interconnects, and DTSP interconnects **150-4** that are non-solder interconnects (e.g., plated interconnects).

(94) FIG. **24A** illustrates an assembly **330** including the package substrate portion **153**. The package substrate portion **153** may be manufactured using any suitable technique, such as a PCB technique or a redistribution layer technique.

(95) FIG. **24B** illustrates an assembly **332** subsequent to forming conductive pillars **134** on the top surface of the package substrate portion **153** of the assembly **330** (FIG. **24A**). The conductive pillars **134** may be disposed around a de-population region **155** in which no conductive pillars **134** are present. The conductive pillars **134** may take the form of any of the embodiments disclosed herein, and may be formed using any suitable technique (e.g., plating). For example, the conductive pillars **134** may include copper.

(96) FIG. **24C** illustrates an assembly **334** subsequent to placing the die **114-1** in the de-population region **155** of the assembly **332** (FIG. **24B**) and coupling the die **114-1** to the package substrate portion **153**. In particular, the conductive contacts **122** at the bottom surface of the die **114-1** may be coupled to the conductive contacts **146** at the top surface of the package substrate portion **153** via DTSP interconnects **150-1**. The DTSP interconnects **150-1** may take any of the forms disclosed herein, such as solder interconnects or anisotropic conductive material interconnects.

(97) FIG. **24D** illustrates an assembly **336** subsequent to providing a mold material **132** around the

die **114-1** and the conductive pillars **134** of the assembly **334** (FIG. **24C**) to complete the package substrate portion **151**. In some embodiments, the mold material **132** may be initially deposited on and over the tops of the conductive pillars **134** and the die **114-1**, then polished back to expose the conductive contacts **122** at the top surface of the die **114-1**, and the top surfaces of the conductive pillars **134**.

(98) FIG. **24E** illustrates an assembly **338** subsequent to forming the package substrate portion **148** on the assembly **336** (FIG. **24D**). The package substrate portion **148** may be formed using any suitable techniques, such as any of the techniques discussed above with reference to the formation of the package substrate portion **115** of FIG. **19G**. In some embodiments, forming the package substrate portion **148** may include plating the conductive contacts **122** of the die **114-1** with a metal or other conductive material as part of forming the proximate conductive contacts **146** of the package substrate **102**; consequently, the DTPS interconnects **150-4** between the die **114-1** and the package substrate portion **148** may be plated interconnects. The dies **114-2**, **114-3**, and **114-4** may then be attached to the top surface of the package substrate portion **148** in accordance with any of the techniques discussed above with reference to FIG. **23B** to form the microelectronic assembly **100** of FIG. **21**.

(99) FIGS. **25A-25F** are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly **100** of FIG. **22**, in accordance with various embodiments. In some embodiments, microelectronic assemblies **100** manufactured in accordance with the process of FIGS. **25A-25F** may have DTPS interconnects **150-1** that are non-solder interconnects (e.g., plated interconnects), and DTPS interconnects **150-4** that are non-solder interconnects (e.g., plated interconnects).

(100) FIG. **25A** illustrates an assembly **340** subsequent to forming a plurality of conductive pillars **134** and a placement ring **136** on a carrier **202**. The conductive pillars **134** may take any of the forms disclosed herein, and may be formed using any suitable technique (e.g., the techniques discussed above with reference to FIG. **24B**). The placement ring **136** may take any of the forms disclosed herein, and may be formed using any suitable technique (e.g., any of the techniques disclosed herein). The placement ring **136** may surround a de-population region **155** in which no conductive pillars **134** are present.

(101) FIG. **25B** illustrates an assembly **342** subsequent to positioning the die **114-1** in the de-population region **155** within the placement ring **136** of the assembly **340** (FIG. **25A**). As noted above, the placement ring **136** may complement the footprint of the die **114-1**, allowing the die **114-1** to be properly positioned.

(102) FIG. **25C** illustrates an assembly **344** subsequent to providing a mold material **132** around the conductive pillars **134** and placement ring **136** of the assembly **342** (FIG. **25B**) to complete the package substrate portion **151**. In some embodiments, the mold material **132** may be initially deposited on and over the tops of the conductive pillars **134** and the die **114-1**, then polished back to expose the conductive contacts **122** at the surface of the die **114-1**, and the surfaces of the conductive pillars **134**.

(103) FIG. **25D** illustrates an assembly **346** subsequent to forming the package substrate portion **153** on the assembly **344** (FIG. **25C**). The package substrate portion **153** may be formed using any suitable techniques, such as any of the techniques discussed above with reference to the formation of the package substrate portion **115** of FIG. **19G**. In some embodiments, forming the package substrate portion **153** may include plating the conductive contacts **122** of the die **114-1** with a metal or other conductive material as part of forming the proximate conductive contacts **146** of the package substrate **102**; consequently, the DTPS interconnects **150-1** between the die **114-1** and the package substrate portion **148** may be plated interconnects.

(104) FIG. **25E** illustrates an assembly **347** subsequent to attaching another carrier **204** to the top surface of the assembly **346** (FIG. **25D**). The carrier **204** may take the form of any of the embodiments of the carrier **202** disclosed herein.

(105) FIG. 25F illustrates an assembly 348 subsequent to removing the carrier 202 from the assembly 347 (FIG. 25E) and flipping the result so that the package substrate portion 151 and the other conductive contacts 122 of the die 114-1 are exposed. The package substrate portion 148 may then be formed on the assembly 348 in accordance with any of the techniques discussed above with reference to FIG. 24E, and the dies 114-2, 114-3, and 114-4 may be attached to the top surface of the package substrate portion 148 (e.g., in accordance with any of the techniques discussed above with reference to FIG. 23B) to form the microelectronic assembly 100 of FIG. 21.

(106) In any of the embodiments disclosed herein, a portion of the package substrate 102 may be formed by assembling two separately manufactured sub-portions. For example, FIGS. 26A-26D are side, cross-sectional views of various stages in another example process for manufacturing the microelectronic assembly 100 of FIG. 21, in accordance with various embodiments. The process of FIGS. 26A-26D includes the assembly of the package substrate portion 153 from two sub-portions, but any package substrate 102 (or portion thereof) may be formed from multiple sub-portions.

(107) FIG. 26A illustrates an assembly 350 subsequent to forming a package substrate sub-portion 153A and forming conductive pillars 134 thereon. The conductive pillars 134 may take the form of any of the embodiments disclosed herein, and the package substrate sub-portion 153A may represent the top half of the package substrate portion 153, as discussed further below.

(108) FIG. 26B illustrates an assembly 352 subsequent to attaching a die 114-1 to the assembly 350 (FIG. 26A), providing a mold material 132 around the conductive pillars 134 and the die 114-1 to complete the package substrate portion 151, and forming a package substrate portion 148 on the package substrate portion 151. These operations may take any of the forms discussed above.

(109) FIG. 26C illustrates an assembly 354 subsequent to bringing the assembly 352 (FIG. 26B) into alignment with a package substrate sub-portion 153B. In particular, the package substrate sub-portion 153A may be brought proximate to the package substrate sub-portion 153B.

(110) FIG. 26D illustrates an assembly 356 subsequent to coupling the package substrate sub-portion 153A and the package substrate sub-portion 153B of the assembly 354 (FIG. 26C) together to form the package substrate portion 153. The dies 114-2, 114-3, and 114-4 may be attached to the top surface of the package substrate portion 148 (e.g., in accordance with any of the techniques discussed above with reference to FIG. 23B, such as solder or anisotropic conductive material techniques) to form the microelectronic assembly 100 of FIG. 21.

(111) The microelectronic assemblies 100 disclosed herein may include conductive pillars 134 in the package substrate 102 even when the die 114-1 is not embedded in the package substrate 102 (e.g., even when no package substrate portion 148 is present). For example, FIG. 27 illustrates an example microelectronic assembly 100 in which the package substrate 102 includes conductive pillars 134 without a package substrate portion 148. In the microelectronic assembly 100 of FIG. 27, the conductive contacts 122 at the bottom surface of the die 114-2 are coupled to the conductive pillars 134 via DTPS interconnects 150-2, and the conductive contacts 124 at the bottom surface of the die 114-2 are coupled to the conductive contacts 122 at the top surface of the die 114-1 via DTD interconnects 130-2. Any of the other microelectronic assemblies 100 disclosed herein may include conductive pillars 134, as appropriate.

(112) The microelectronic assemblies 100 disclosed herein may be used for any suitable application. For example, in some embodiments, a microelectronic assembly 100 may be used to provide an ultra-high density and high bandwidth interconnect for field programmable gate array (FPGA) transceivers and III-V amplifiers. For example, the die 114-1 may include FPGA transceiver circuitry or III-V amplifiers, and the die 114-2 may include FPGA logic.

Communications between the die 114-1 and the die 114-2 may experience less delay than if such communications were routed through an intermediate device (e.g., a separate silicon bridge). In some embodiments, the pitch of the DTD interconnects 130-1 between the die 114-1 and the die 114-2 may be less than 100 microns (e.g., between 25 microns and 55 microns) and the pitch of the DTPS interconnects 150-2 between the die 114-2 and the package substrate 102 may be greater

than 80 microns (e.g., between 100 and 150 microns). Such applications may be particularly suitable for military electronics, 5G wireless communications, WiGig communications, and/or millimeter wave communications.

(113) More generally, the microelectronic assemblies **100** disclosed herein may allow “blocks” of different kinds of functional circuits to be distributed into different ones of the dies **114**, instead of having all of the circuits included in a single large die, per some conventional approaches. In some such conventional approaches, a single large die would include all of these different circuits to achieve high bandwidth, low loss communication between the circuits, and some or all of these circuits may be selectively disabled to adjust the capabilities of the large die. However, because the DTD interconnects **130** of the microelectronic assemblies **100** may allow high bandwidth, low loss communication between different ones of the dies **114**, different circuits may be distributed into different dies **114**, reducing the total cost of manufacture, improving yield, and increasing design flexibility by allowing different dies **114** (e.g., dies **114** formed using different fabrication technologies) to be readily swapped to achieve different functionality. Additionally, a die **114** stacked on top of another die **114** may be closer to the heat spreader **131** than if the circuitry of the two dies were combined into a single die farther from the heat spreader **131**, improving thermal performance.

(114) In another example, a die **114-1** that includes active circuitry in a microelectronic assembly **100** may be used to provide an “active” bridge between other dies **114** (e.g., between the dies **114-2** and **114-3**, or between multiple different dies **114-2**, in various embodiments). In some such embodiments, power delivery may be provided to the “bottoms” of the die **114-1** and the other dies **114** through the package substrate **102** without requiring additional layers of package substrate **102** above the die **140-1** through which to route power.

(115) In another example, the die **114-1** in a microelectronic assembly **100** may be a processing device (e.g., a central processing unit, a graphics processing unit, a FPGA, a modem, an applications processor, etc.), and the die **114-2** may include high bandwidth memory, transceiver circuitry, and/or input/output circuitry (e.g., Double Data Rate transfer circuitry, Peripheral Component Interconnect Express circuitry, etc.). In some embodiments, the die **114-1** may include a set of conductive contacts **124** to interface with a high bandwidth memory die **114-2**, a different set of conductive contacts **124** to interface with an input/output circuitry die **114-2**, etc. The particular high bandwidth memory die **114-2**, input/output circuitry die **114-2**, etc. may be selected for the application at hand.

(116) In another example, the die **114-1** in a microelectronic assembly **100** may be a cache memory (e.g., a third level cache memory), and one or more dies **114-2** may be processing devices (e.g., a central processing unit, a graphics processing unit, a FPGA, a modem, an applications processor, etc.) that share the cache memory of the die **114-1**.

(117) The microelectronic assemblies **100** disclosed herein may be included in any suitable electronic component. FIGS. **28-31** illustrate various examples of apparatuses that may include, or be included in, any of the microelectronic assemblies **100** disclosed herein.

(118) FIG. **28** is a top view of a wafer **1500** and dies **1502** that may be included in any of the microelectronic assemblies **100** disclosed herein (e.g., as any suitable ones of the dies **114**). The wafer **1500** may be composed of semiconductor material and may include one or more dies **1502** having IC structures formed on a surface of the wafer **1500**. Each of the dies **1502** may be a repeating unit of a semiconductor product that includes any suitable IC. After the fabrication of the semiconductor product is complete, the wafer **1500** may undergo a singulation process in which the dies **1502** are separated from one another to provide discrete “chips” of the semiconductor product. The die **1502** may be any of the dies **114** disclosed herein. The die **1502** may include one or more transistors (e.g., some of the transistors **1640** of FIG. **29**, discussed below), supporting circuitry to route electrical signals to the transistors, passive components (e.g., signal traces, resistors, capacitors, or inductors), and/or any other IC components. In some embodiments, the wafer **1500**

or the die **1502** may include a memory device (e.g., a random access memory (RAM) device, such as a static RAM (SRAM) device, a magnetic RAM (MRAM) device, a resistive RAM (RRAM) device, a conductive-bridging RAM (CBRAM) device, etc.), a logic device (e.g., an AND, OR, NAND, or NOR gate), or any other suitable circuit element. Multiple ones of these devices may be combined on a single die **1502**. For example, a memory array formed by multiple memory devices may be formed on a same die **1502** as a processing device (e.g., the processing device **1802** of FIG. **31**) or other logic that is configured to store information in the memory devices or execute instructions stored in the memory array. Various ones of the microelectronic assemblies **100** disclosed herein may be manufactured using a die-to-wafer assembly technique in which some dies **114** are attached to a wafer **1500** that include others of the dies **114**, and the wafer **1500** is subsequently singulated.

(119) FIG. **29** is a cross-sectional side view of an IC device **1600** that may be included in any of the microelectronic assemblies **100** disclosed herein (e.g., in any of the dies **114**). One or more of the IC devices **1600** may be included in one or more dies **1502** (FIG. **28**). The IC device **1600** may be formed on a die substrate **1602** (e.g., the wafer **1500** of FIG. **28**) and may be included in a die (e.g., the die **1502** of FIG. **28**). The die substrate **1602** may be a semiconductor substrate composed of semiconductor material systems including, for example, n-type or p-type materials systems (or a combination of both). The die substrate **1602** may include, for example, a crystalline substrate formed using a bulk silicon or a silicon-on-insulator (SOI) substructure. In some embodiments, the die substrate **1602** may be formed using alternative materials, which may or may not be combined with silicon, that include, but are not limited to, germanium, indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide, or gallium antimonide. Further materials classified as group II-VI, III-V, or IV may also be used to form the die substrate **1602**. Although a few examples of materials from which the die substrate **1602** may be formed are described here, any material that may serve as a foundation for an IC device **1600** may be used. The die substrate **1602** may be part of a singulated die (e.g., the dies **1502** of FIG. **28**) or a wafer (e.g., the wafer **1500** of FIG. **28**).

(120) The IC device **1600** may include one or more device layers **1604** disposed on the die substrate **1602**. The device layer **1604** may include features of one or more transistors **1640** (e.g., metal oxide semiconductor field-effect transistors (MOSFETs)) formed on the die substrate **1602**. The device layer **1604** may include, for example, one or more source and/or drain (S/D) regions **1620**, a gate **1622** to control current flow in the transistors **1640** between the S/D regions **1620**, and one or more S/D contacts **1624** to route electrical signals to/from the S/D regions **1620**. The transistors **1640** may include additional features not depicted for the sake of clarity, such as device isolation regions, gate contacts, and the like. The transistors **1640** are not limited to the type and configuration depicted in FIG. **29** and may include a wide variety of other types and configurations such as, for example, planar transistors, non-planar transistors, or a combination of both. Non-planar transistors may include FinFET transistors, such as double-gate transistors or tri-gate transistors, and wrap-around or all-around gate transistors, such as nanoribbon and nanowire transistors.

(121) Each transistor **1640** may include a gate **1622** formed of at least two layers, a gate dielectric and a gate electrode. The gate dielectric may include one layer or a stack of layers. The one or more layers may include silicon oxide, silicon dioxide, silicon carbide, and/or a high-k dielectric material. The high-k dielectric material may include elements such as hafnium, silicon, oxygen, titanium, tantalum, lanthanum, aluminum, zirconium, barium, strontium, yttrium, lead, scandium, niobium, and zinc. Examples of high-k materials that may be used in the gate dielectric include, but are not limited to, hafnium oxide, hafnium silicon oxide, lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, tantalum oxide, titanium oxide, barium strontium titanium oxide, barium titanium oxide, strontium titanium oxide, yttrium oxide, aluminum oxide, lead scandium tantalum oxide, and lead zinc niobate. In some embodiments, an annealing process



may be carried out on the gate dielectric to improve its quality when a high-k material is used.

(122) The gate electrode may be formed on the gate dielectric and may include at least one p-type work function metal or n-type work function metal, depending on whether the transistor **1640** is to be a p-type metal oxide semiconductor (PMOS) or an n-type metal oxide semiconductor (NMOS) transistor. In some implementations, the gate electrode may consist of a stack of two or more metal layers, where one or more metal layers are work function metal layers and at least one metal layer is a fill metal layer. Further metal layers may be included for other purposes, such as a barrier layer. For a PMOS transistor, metals that may be used for the gate electrode include, but are not limited to, ruthenium, palladium, platinum, cobalt, nickel, conductive metal oxides (e.g., ruthenium oxide), and any of the metals discussed below with reference to an NMOS transistor (e.g., for work function tuning). For an NMOS transistor, metals that may be used for the gate electrode include, but are not limited to, hafnium, zirconium, titanium, tantalum, aluminum, alloys of these metals, carbides of these metals (e.g., hafnium carbide, zirconium carbide, titanium carbide, tantalum carbide, and aluminum carbide), and any of the metals discussed above with reference to a PMOS transistor (e.g., for work function tuning).

(123) In some embodiments, when viewed as a cross-section of the transistor **1640** along the source-channel-drain direction, the gate electrode may consist of a U-shaped structure that includes a bottom portion substantially parallel to the surface of the die substrate **1602** and two sidewall portions that are substantially perpendicular to the top surface of the die substrate **1602**. In other embodiments, at least one of the metal layers that form the gate electrode may simply be a planar layer that is substantially parallel to the top surface of the die substrate **1602** and does not include sidewall portions substantially perpendicular to the top surface of the die substrate **1602**. In other embodiments, the gate electrode may consist of a combination of U-shaped structures and planar, non-U-shaped structures. For example, the gate electrode may consist of one or more U-shaped metal layers formed atop one or more planar, non-U-shaped layers.

(124) In some embodiments, a pair of sidewall spacers may be formed on opposing sides of the gate stack to bracket the gate stack. The sidewall spacers may be formed from materials such as silicon nitride, silicon oxide, silicon carbide, silicon nitride doped with carbon, and silicon oxynitride. Processes for forming sidewall spacers are well known in the art and generally include deposition and etching process steps. In some embodiments, a plurality of spacer pairs may be used; for instance, two pairs, three pairs, or four pairs of sidewall spacers may be formed on opposing sides of the gate stack.

(125) The S/D regions **1620** may be formed within the die substrate **1602** adjacent to the gate **1622** of each transistor **1640**. The S/D regions **1620** may be formed using an implantation/diffusion process or an etching/deposition process, for example. In the former process, dopants such as boron, aluminum, antimony, phosphorous, or arsenic may be ion-implanted into the die substrate **1602** to form the S/D regions **1620**. An annealing process that activates the dopants and causes them to diffuse farther into the die substrate **1602** may follow the ion-implantation process. In the latter process, the die substrate **1602** may first be etched to form recesses at the locations of the S/D regions **1620**. An epitaxial deposition process may then be carried out to fill the recesses with material that is used to fabricate the S/D regions **1620**. In some implementations, the S/D regions **1620** may be fabricated using a silicon alloy such as silicon germanium or silicon carbide. In some embodiments, the epitaxially deposited silicon alloy may be doped in situ with dopants such as boron, arsenic, or phosphorous. In some embodiments, the S/D regions **1620** may be formed using one or more alternate semiconductor materials such as germanium or a group III-V material or alloy. In further embodiments, one or more layers of metal and/or metal alloys may be used to form the S/D regions **1620**.

(126) Electrical signals, such as power and/or input/output (I/O) signals, may be routed to and/or from the devices (e.g., transistors **1640**) of the device layer **1604** through one or more interconnect layers disposed on the device layer **1604** (illustrated in FIG. 29 as interconnect layers **1606-1610**).

For example, electrically conductive features of the device layer **1604** (e.g., the gate **1622** and the S/D contacts **1624**) may be electrically coupled with the interconnect structures **1628** of the interconnect layers **1606-1610**. The one or more interconnect layers **1606-1610** may form a metallization stack (also referred to as an “ILD stack”) **1619** of the IC device **1600**.

(127) The interconnect structures **1628** may be arranged within the interconnect layers **1606-1610** to route electrical signals according to a wide variety of designs; in particular, the arrangement is not limited to the particular configuration of interconnect structures **1628** depicted in FIG. 29. Although a particular number of interconnect layers **1606-1610** is depicted in FIG. 29, embodiments of the present disclosure include IC devices having more or fewer interconnect layers than depicted.

(128) In some embodiments, the interconnect structures **1628** may include lines **1628a** and/or vias **1628b** filled with an electrically conductive material such as a metal. The lines **1628a** may be arranged to route electrical signals in a direction of a plane that is substantially parallel with a surface of the die substrate **1602** upon which the device layer **1604** is formed. For example, the lines **1628a** may route electrical signals in a direction in and out of the page from the perspective of FIG. 29. The vias **1628b** may be arranged to route electrical signals in a direction of a plane that is substantially perpendicular to the surface of the die substrate **1602** upon which the device layer **1604** is formed. In some embodiments, the vias **1628b** may electrically couple lines **1628a** of different interconnect layers **1606-1610** together.

(129) The interconnect layers **1606-1610** may include a dielectric material **1626** disposed between the interconnect structures **1628**, as shown in FIG. 29. In some embodiments, the dielectric material **1626** disposed between the interconnect structures **1628** in different ones of the interconnect layers **1606-1610** may have different compositions; in other embodiments, the composition of the dielectric material **1626** between different interconnect layers **1606-1610** may be the same.

(130) A first interconnect layer **1606** (referred to as Metal 1 or “M1”) may be formed directly on the device layer **1604**. In some embodiments, the first interconnect layer **1606** may include lines **1628a** and/or vias **1628b**, as shown. The lines **1628a** of the first interconnect layer **1606** may be coupled with contacts (e.g., the S/D contacts **1624**) of the device layer **1604**.

(131) A second interconnect layer **1608** (referred to as Metal 2 or “M2”) may be formed directly on the first interconnect layer **1606**. In some embodiments, the second interconnect layer **1608** may include vias **1628b** to couple the lines **1628a** of the second interconnect layer **1608** with the lines **1628a** of the first interconnect layer **1606**. Although the lines **1628a** and the vias **1628b** are structurally delineated with a line within each interconnect layer (e.g., within the second interconnect layer **1608**) for the sake of clarity, the lines **1628a** and the vias **1628b** may be structurally and/or materially contiguous (e.g., simultaneously filled during a dual-damascene process) in some embodiments.

(132) A third interconnect layer **1610** (referred to as Metal 3 or “M3”) (and additional interconnect layers, as desired) may be formed in succession on the second interconnect layer **1608** according to similar techniques and configurations described in connection with the second interconnect layer **1608** or the first interconnect layer **1606**. In some embodiments, the interconnect layers that are “higher up” in the metallization stack **1619** in the IC device **1600** (i.e., farther away from the device layer **1604**) may be thicker.

(133) The IC device **1600** may include a solder resist material **1634** (e.g., polyimide or similar material) and one or more conductive contacts **1636** formed on the interconnect layers **1606-1610**. In FIG. 29, the conductive contacts **1636** are illustrated as taking the form of bond pads. The conductive contacts **1636** may be electrically coupled with the interconnect structures **1628** and configured to route the electrical signals of the transistor(s) **1640** to other external devices. For example, solder bonds may be formed on the one or more conductive contacts **1636** to mechanically and/or electrically couple a chip including the IC device **1600** with another

component (e.g., a circuit board). The IC device **1600** may include additional or alternate structures to route the electrical signals from the interconnect layers **1606-1610**; for example, the conductive contacts **1636** may include other analogous features (e.g., posts) that route the electrical signals to external components. The conductive contacts **1636** may serve as the conductive contacts **122** or **124**, as appropriate.

(134) In some embodiments in which the IC device **1600** is a double-sided die (e.g., like the die **114-1**), the IC device **1600** may include another metallization stack (not shown) on the opposite side of the device layer(s) **1604**. This metallization stack may include multiple interconnect layers as discussed above with reference to the interconnect layers **1606-1610**, to provide conductive pathways (e.g., including conductive lines and vias) between the device layer(s) **1604** and additional conductive contacts (not shown) on the opposite side of the IC device **1600** from the conductive contacts **1636**. These additional conductive contacts may serve as the conductive contacts **122** or **124**, as appropriate.

(135) In other embodiments in which the IC device **1600** is a double-sided die (e.g., like the die **114-1**), the IC device **1600** may include one or more TSVs through the die substrate **1602**; these TSVs may make contact with the device layer(s) **1604**, and may provide conductive pathways between the device layer(s) **1604** and additional conductive contacts (not shown) on the opposite side of the IC device **1600** from the conductive contacts **1636**. These additional conductive contacts may serve as the conductive contacts **122** or **124**, as appropriate.

(136) FIG. **30** is a cross-sectional side view of an IC device assembly **1700** that may include any of the microelectronic assemblies **100** disclosed herein. In some embodiments, the IC device assembly **1700** may be a microelectronic assembly **100**. The IC device assembly **1700** includes a number of components disposed on a circuit board **1702** (which may be, e.g., a motherboard). The IC device assembly **1700** includes components disposed on a first face **1740** of the circuit board **1702** and an opposing second face **1742** of the circuit board **1702**; generally, components may be disposed on one or both faces **1740** and **1742**. Any of the IC packages discussed below with reference to the IC device assembly **1700** may take the form of any suitable ones of the embodiments of the microelectronic assemblies **100** disclosed herein.

(137) In some embodiments, the circuit board **1702** may be a PCB including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. Any one or more of the metal layers may be formed in a desired circuit pattern to route electrical signals (optionally in conjunction with other metal layers) between the components coupled to the circuit board **1702**. In other embodiments, the circuit board **1702** may be a non-PCB substrate. In some embodiments the circuit board **1702** may be, for example, the circuit board **133**.

(138) The IC device assembly **1700** illustrated in FIG. **30** includes a package-on-interposer structure **1736** coupled to the first face **1740** of the circuit board **1702** by coupling components **1716**. The coupling components **1716** may electrically and mechanically couple the package-on-interposer structure **1736** to the circuit board **1702**, and may include solder balls (as shown in FIG. **30**), male and female portions of a socket, an adhesive, an underfill material, and/or any other suitable electrical and/or mechanical coupling structure.

(139) The package-on-interposer structure **1736** may include an IC package **1720** coupled to an interposer **1704** by coupling components **1718**. The coupling components **1718** may take any suitable form for the application, such as the forms discussed above with reference to the coupling components **1716**. Although a single IC package **1720** is shown in FIG. **30**, multiple IC packages may be coupled to the interposer **1704**; indeed, additional interposers may be coupled to the interposer **1704**. The interposer **1704** may provide an intervening substrate used to bridge the circuit board **1702** and the IC package **1720**. The IC package **1720** may be or include, for example, a die (the die **1502** of FIG. **28**), an IC device (e.g., the IC device **1600** of FIG. **29**), or any other suitable component. Generally, the interposer **1704** may spread a connection to a wider pitch or reroute a connection to a different connection. For example, the interposer **1704** may couple the IC

package **1720** (e.g., a die) to a set of ball grid array (BGA) conductive contacts of the coupling components **1716** for coupling to the circuit board **1702**. In the embodiment illustrated in FIG. **30**, the IC package **1720** and the circuit board **1702** are attached to opposing sides of the interposer **1704**; in other embodiments, the IC package **1720** and the circuit board **1702** may be attached to a same side of the interposer **1704**. In some embodiments, three or more components may be interconnected by way of the interposer **1704**.

(140) In some embodiments, the interposer **1704** may be formed as a PCB, including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. In some embodiments, the interposer **1704** may be formed of an epoxy resin, a fiberglass-reinforced epoxy resin, an epoxy resin with inorganic fillers, a ceramic material, or a polymer material such as polyimide. In some embodiments, the interposer **1704** may be formed of alternate rigid or flexible materials that may include the same materials described above for use in a semiconductor substrate, such as silicon, germanium, and other group III-V and group IV materials. The interposer **1704** may include metal interconnects **1708** and vias **1710**, including but not limited to TSVs **1706**. The interposer **1704** may further include embedded devices **1714**, including both passive and active devices. Such devices may include, but are not limited to, capacitors, decoupling capacitors, resistors, inductors, fuses, diodes, transformers, sensors, electrostatic discharge (ESD) devices, and memory devices. More complex devices such as radio frequency devices, power amplifiers, power management devices, antennas, arrays, sensors, and microelectromechanical systems (MEMS) devices may also be formed on the interposer **1704**. The package-on-interposer structure **1736** may take the form of any of the package-on-interposer structures known in the art.

(141) The IC device assembly **1700** may include an IC package **1724** coupled to the first face **1740** of the circuit board **1702** by coupling components **1722**. The coupling components **1722** may take the form of any of the embodiments discussed above with reference to the coupling components **1716**, and the IC package **1724** may take the form of any of the embodiments discussed above with reference to the IC package **1720**.

(142) The IC device assembly **1700** illustrated in FIG. **30** includes a package-on-package structure **1734** coupled to the second face **1742** of the circuit board **1702** by coupling components **1728**. The package-on-package structure **1734** may include an IC package **1726** and an IC package **1732** coupled together by coupling components **1730** such that the IC package **1726** is disposed between the circuit board **1702** and the IC package **1732**. The coupling components **1728** and **1730** may take the form of any of the embodiments of the coupling components **1716** discussed above, and the IC packages **1726** and **1732** may take the form of any of the embodiments of the IC package **1720** discussed above. The package-on-package structure **1734** may be configured in accordance with any of the package-on-package structures known in the art.

(143) FIG. **31** is a block diagram of an example electrical device **1800** that may include one or more of the microelectronic assemblies **100** disclosed herein. For example, any suitable ones of the components of the electrical device **1800** may include one or more of the IC device assemblies **1700**, IC devices **1600**, or dies **1502** disclosed herein, and may be arranged in any of the microelectronic assemblies **100** disclosed herein. A number of components are illustrated in FIG. **31** as included in the electrical device **1800**, but any one or more of these components may be omitted or duplicated, as suitable for the application. In some embodiments, some or all of the components included in the electrical device **1800** may be attached to one or more motherboards. In some embodiments, some or all of these components are fabricated onto a single system-on-a-chip (SoC) die.

(144) Additionally, in various embodiments, the electrical device **1800** may not include one or more of the components illustrated in FIG. **31**, but the electrical device **1800** may include interface circuitry for coupling to the one or more components. For example, the electrical device **1800** may not include a display device **1806**, but may include display device interface circuitry (e.g., a

connector and driver circuitry) to which a display device **1806** may be coupled. In another set of examples, the electrical device **1800** may not include an audio input device **1824** or an audio output device **1808**, but may include audio input or output device interface circuitry (e.g., connectors and supporting circuitry) to which an audio input device **1824** or audio output device **1808** may be coupled.

(145) The electrical device **1800** may include a processing device **1802** (e.g., one or more processing devices). As used herein, the term “processing device” or “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory. The processing device **1802** may include one or more digital signal processors (DSPs), application-specific integrated circuits (ASICs), central processing units (CPUs), graphics processing units (GPUs), cryptoprocessors (specialized processors that execute cryptographic algorithms within hardware), server processors, or any other suitable processing devices. The electrical device **1800** may include a memory **1804**, which may itself include one or more memory devices such as volatile memory (e.g., dynamic random access memory (DRAM)), nonvolatile memory (e.g., read-only memory (ROM)), flash memory, solid state memory, and/or a hard drive. In some embodiments, the memory **1804** may include memory that shares a die with the processing device **1802**. This memory may be used as cache memory and may include embedded dynamic random access memory (eDRAM) or spin transfer torque magnetic random access memory (STT-MRAM).

(146) In some embodiments, the electrical device **1800** may include a communication chip **1812** (e.g., one or more communication chips). For example, the communication chip **1812** may be configured for managing wireless communications for the transfer of data to and from the electrical device **1800**. The term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a nonsolid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not.

(147) The communication chip **1812** may implement any of a number of wireless standards or protocols, including but not limited to Institute for Electrical and Electronic Engineers (IEEE) standards including Wi-Fi (IEEE 802.11 family), IEEE 802.16 standards (e.g., IEEE 802.16-2005 Amendment), Long-Term Evolution (LTE) project along with any amendments, updates, and/or revisions (e.g., advanced LTE project, ultra mobile broadband (UMB) project (also referred to as “3GPP2”), etc.). IEEE 802.16 compatible Broadband Wireless Access (BWA) networks are generally referred to as WiMAX networks, an acronym that stands for Worldwide Interoperability for Microwave Access, which is a certification mark for products that pass conformity and interoperability tests for the IEEE 802.16 standards. The communication chip **1812** may operate in accordance with a Global System for Mobile Communication (GSM), General Packet Radio Service (GPRS), Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Evolved HSPA (E-HSPA), or LTE network. The communication chip **1812** may operate in accordance with Enhanced Data for GSM Evolution (EDGE), GSM EDGE Radio Access Network (GERAN), Universal Terrestrial Radio Access Network (UTRAN), or Evolved UTRAN (E-UTRAN). The communication chip **1812** may operate in accordance with Code Division Multiple Access (CDMA), Time Division Multiple Access (TDMA), Digital Enhanced Cordless Telecommunications (DECT), Evolution-Data Optimized (EV-DO), and derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The communication chip **1812** may operate in accordance with other wireless protocols in other embodiments. The electrical device **1800** may include an antenna **1822** to facilitate wireless communications and/or to receive other wireless communications (such as AM or FM radio transmissions).

(148) In some embodiments, the communication chip **1812** may manage wired communications,

such as electrical, optical, or any other suitable communication protocols (e.g., the Ethernet). As noted above, the communication chip **1812** may include multiple communication chips. For instance, a first communication chip **1812** may be dedicated to shorter-range wireless communications such as Wi-Fi or Bluetooth, and a second communication chip **1812** may be dedicated to longer-range wireless communications such as global positioning system (GPS), EDGE, GPRS, CDMA, WiMAX, LTE, EV-DO, or others. In some embodiments, a first communication chip **1812** may be dedicated to wireless communications, and a second communication chip **1812** may be dedicated to wired communications.

(149) The electrical device **1800** may include battery/power circuitry **1814**. The battery/power circuitry **1814** may include one or more energy storage devices (e.g., batteries or capacitors) and/or circuitry for coupling components of the electrical device **1800** to an energy source separate from the electrical device **1800** (e.g., AC line power).

(150) The electrical device **1800** may include a display device **1806** (or corresponding interface circuitry, as discussed above). The display device **1806** may include any visual indicators, such as a heads-up display, a computer monitor, a projector, a touchscreen display, a liquid crystal display (LCD), a light-emitting diode display, or a flat panel display.

(151) The electrical device **1800** may include an audio output device **1808** (or corresponding interface circuitry, as discussed above). The audio output device **1808** may include any device that generates an audible indicator, such as speakers, headsets, or earbuds.

(152) The electrical device **1800** may include an audio input device **1824** (or corresponding interface circuitry, as discussed above). The audio input device **1824** may include any device that generates a signal representative of a sound, such as microphones, microphone arrays, or digital instruments (e.g., instruments having a musical instrument digital interface (MIDI) output).

(153) The electrical device **1800** may include a GPS device **1818** (or corresponding interface circuitry, as discussed above). The GPS device **1818** may be in communication with a satellite-based system and may receive a location of the electrical device **1800**, as known in the art.

(154) The electrical device **1800** may include an other output device **1810** (or corresponding interface circuitry, as discussed above). Examples of the other output device **1810** may include an audio codec, a video codec, a printer, a wired or wireless transmitter for providing information to other devices, or an additional storage device.

(155) The electrical device **1800** may include an other input device **1820** (or corresponding interface circuitry, as discussed above). Examples of the other input device **1820** may include an accelerometer, a gyroscope, a compass, an image capture device, a keyboard, a cursor control device such as a mouse, a stylus, a touchpad, a bar code reader, a Quick Response (QR) code reader, any sensor, or a radio frequency identification (RFID) reader.

(156) The electrical device **1800** may have any desired form factor, such as a hand-held or mobile electrical device (e.g., a cell phone, a smart phone, a mobile internet device, a music player, a tablet computer, a laptop computer, a netbook computer, an ultrabook computer, a personal digital assistant (PDA), an ultra mobile personal computer, etc.), a desktop electrical device, a server or other networked computing component, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a vehicle control unit, a digital camera, a digital video recorder, or a wearable electrical device. In some embodiments, the electrical device **1800** may be any other electronic device that processes data.

(157) The following paragraphs provide various examples of the embodiments disclosed herein.

(158) Example 1 is a method of manufacturing a microelectronic assembly, including: forming first interconnects between a first die and a second die, wherein the first die has a first surface and an opposing second surface, the first die includes first conductive contacts at its first surface and second conductive contacts at its first surface, the second die has a first surface and an opposing second surface, the second die includes first conductive contacts at its first surface and second conductive contacts at its second surface, and the first interconnects couple the first conductive

contacts of the first die with the second conductive contacts of the second die; and after forming the first interconnects, forming second interconnects between the first conductive contacts of the second die and first conductive contacts of a package substrate, and forming third interconnects between the second conductive contacts of the first die and second conductive contacts of the package substrate.

(159) Example 2 may include the subject matter of Example 1, and may further include forming the package substrate.

(160) Example 3 may include the subject matter of Example 2, and may further specify that forming the package substrate includes: forming an initial package substrate; and forming a recess in the initial package substrate; wherein the first conductive contacts of the package substrate are at a bottom of the recess.

(161) Example 4 may include the subject matter of Example 3, and may further specify that the recess is a first recess, and forming the package substrate further includes: forming a second recess in the initial package substrate; wherein the second conductive contacts of the package substrate are at a bottom of the second recess.

(162) Example 5 may include the subject matter of any of Examples 1-4, and may further include:

(163) before forming the first interconnects, placing the first die on a carrier, wherein the second surface of the first die is between the carrier and the first surface of the first die; and

(164) after forming the third interconnects, removing the carrier.

(165) Example 6 may include the subject matter of any of Examples 1-5, and may further specify that the first interconnects do not include solder.

(166) Example 7 may include the subject matter of any of Examples 1-6, and may further specify that the first interconnects are copper-to-copper interconnects.

(167) Example 8 may include the subject matter of any of Examples 1-7, and may further specify that the first interconnects include an anisotropic conductive material.

(168) Example 9 may include the subject matter of any of Examples 1-8, and may further specify that the second interconnects include solder.

(169) Example 10 may include the subject matter of any of Examples 1-9, and may further specify that the third interconnects include solder.

(170) Example 11 may include the subject matter of any of Examples 1-10, and may further specify that a pitch of the first interconnects is less than a pitch of the third interconnects.

(171) Example 12 may include the subject matter of any of Examples 1-11, and may further include attaching the package substrate to a circuit board.

(172) Example 13 is a method of manufacturing a microelectronic assembly, including: forming first interconnects between a first die and a package substrate, wherein the first die has a first surface and an opposing second surface, the first die includes first conductive contacts at its first surface and second conductive contacts at its second surface, the first surface of the first die is between the package substrate and the second surface of the first die, and the first interconnects are formed between the first conductive contacts of the first die and first conductive contacts of the package substrate; after forming the first interconnects, forming second interconnects between the first die and a second die, wherein the second die has a first surface and an opposing second surface, the second die includes first conductive contacts at its first surface and second conductive contacts at its second surface, and the second interconnects couple the second conductive contacts of the first die with the first conductive contacts of the second die; and after forming the first interconnects, forming third interconnects between the second conductive contacts of the second die and the second conductive contacts of the package substrate.

(173) Example 14 may include the subject matter of Example 13, and may further include forming the package substrate.

(174) Example 15 may include the subject matter of Example 14, and may further specify that forming the package substrate includes: forming an initial package substrate; and forming a recess

in the initial package substrate;

(175) wherein the first conductive contacts of the package substrate are at a bottom of the recess.

(176) Example 16 may include the subject matter of Example 15, and may further specify that the recess is a first recess, and forming the package substrate further includes: forming a second recess in the initial package substrate; wherein the second conductive contacts of the package substrate are at a bottom of the second recess.

(177) Example 17 may include the subject matter of any of Examples 13-16, and may further specify that the first interconnects include solder.

(178) Example 18 may include the subject matter of any of Examples 13-17, and may further specify that the first interconnects include an anisotropic conductive material.

(179) Example 19 may include the subject matter of any of Examples 13-18, and may further specify that the second interconnects include solder.

(180) Example 20 may include the subject matter of any of Examples 13-19, and may further specify that the third interconnects include solder.

(181) Example 21 may include the subject matter of any of Examples 13-20, and may further specify that a pitch of the second interconnects is less than a pitch of the first interconnects.

(182) Example 22 may include the subject matter of any of Examples 13-21, and may further specify that a distance between the second die and a proximate surface of the package substrate is greater than a distance between the first die and a proximate surface of the package substrate.

(183) Example 23 is a method of manufacturing a microelectronic assembly, including: forming interconnects between a die and a package substrate, wherein the die has a first surface and an opposing second surface, the die includes first conductive contacts at its first surface and second conductive contacts at its second surface, the first surface of the die is between the package substrate and the second surface of the die, and the interconnects are formed between the first conductive contacts of the die and conductive contacts of the package substrate; and providing a mold material on at least a portion of the package substrate.

(184) Example 24 may include the subject matter of Example 23, and may further specify that the die is a first die, the interconnects are first interconnects, the conductive contacts of the package substrate are first conductive contacts, and the method further includes: after forming the first interconnects, forming second interconnects between the first die and a second die, wherein the second die has a first surface and an opposing second surface, the second die includes first conductive contacts at its first surface and second conductive contacts at its first surface, and the second interconnects couple the second conductive contacts of the first die with the first conductive contacts of the second die; and after forming the first interconnects, forming third interconnects between the second conductive contacts of the second die and the second conductive contacts of the package substrate.

(185) Example 25 may include the subject matter of Example 24, and may further specify that the second interconnects include solder.

(186) Example 26 may include the subject matter of any of Examples 24-25, and may further specify that the third interconnects include solder.

(187) Example 27 may include the subject matter of any of Examples 24-26, and may further specify that a pitch of the second interconnects is less than a pitch of the first interconnects.

(188) Example 28 may include the subject matter of any of Examples 24-27, and may further specify that a distance between the second die and a proximate surface of the package substrate is greater than a distance between the first die and a proximate surface of the package substrate.

(189) Example 29 may include the subject matter of any of Examples 23-28, and may further include forming the package substrate.

(190) Example 30 may include the subject matter of Example 29, and may further specify that forming the package substrate includes: forming an initial package substrate; and forming a recess in the initial package substrate; wherein the conductive contacts of the package substrate are at a



bottom of the recess.

(191) Example 31 may include the subject matter of any of Examples 23-30, and may further include attaching the package substrate to a circuit board.

(192) Example 32 is a method of manufacturing a microelectronic assembly, including: forming a portion of a package substrate, wherein the portion of the package substrate includes a cavity; placing a first die in the cavity, wherein the first die has a first surface and an opposing second surface, and the first die includes first conductive contacts at its first surface and second conductive contacts at its second surface; after placing the first die in the cavity, forming first interconnects between a second die and the portion of the package substrate, wherein the second die has a first surface and an opposing second surface, the second die includes first conductive contacts at its first surface and second conductive contacts at its second surface, the first surface of the second die is between the package substrate and the second surface of the second die, and the first interconnects are formed between the first conductive contacts of the second die and first conductive contacts of the portion of the package substrate; and after placing the first die in the cavity, forming second interconnects between the second die and the first die, wherein the second interconnects are formed between the second conductive contacts of the first die and the second conductive contacts of the second die.

(193) Example 33 may include the subject matter of Example 32, and may further specify that the portion of the package substrate is a first portion, and the method further includes forming a second portion of the package substrate proximate to the first surface of the first die, wherein forming the second portion of the package substrate includes forming third interconnects between the first die and the second portion of the package substrate, and the third interconnects are formed between the first conductive contacts of the first die and second conductive contacts of the second portion of the package substrate.

(194) Example 34 may include the subject matter of Example 33, and may further specify that the third interconnects include metal-to-metal interconnects.

(195) Example 35 may include the subject matter of any of Examples 32-34, and may further specify that the first interconnects include solder.

(196) Example 36 may include the subject matter of any of Examples 32-35, and may further specify that the first interconnects include an anisotropic conductive material.

(197) Example 37 may include the subject matter of any of Examples 32-36, and may further include, after placing the first die in the cavity, providing a mold material around the first die in the cavity.

## Claims

1. A microelectronic assembly, comprising: a multi-layer structure including multiple layers of dielectric material, wherein the multi-layer structure includes one or more conductive pathways through the dielectric material; a first die, wherein the first die is conductively coupled to the multi-layer structure; a second die; wherein: the first die is at least partially between the second die and the multi-layer structure, the first die is bonded to the second die by die-to-die interconnects having a pitch between 7 microns and 100 microns, and the microelectronic assembly includes interconnects between the second die and the multi-layer structure, wherein the interconnects do not extend through the first die; and a single mold material over the first die and the second die, the single mold material in contact with a sidewall of the first die, the single mold material having an uppermost surface above an uppermost surface of the second die, and the single mold material on and in contact with the uppermost surface of the second die.

2. The microelectronic assembly of claim 1, wherein the multi-layer structure includes one or more redistribution layers.

3. The microelectronic assembly of claim 1, wherein a footprint of the first die is contained within

a footprint of the second die.

4. The microelectronic assembly of claim 1, wherein the first die includes through-substrate vias (TSVs) through a semiconductor substrate of the first die.

5. The microelectronic assembly of claim 1, wherein the first die is conductively coupled to the multi-layer structure by solder.

6. The microelectronic assembly of claim 1, wherein the single mold material is between a portion of the second die and the multi-layer structure.

7. The microelectronic assembly of claim 6, wherein the interconnects extend through the single mold material.

8. The microelectronic assembly of claim 1, wherein a line from the second die to the multi-layer structure, perpendicular to a plane of a layer of dielectric material of the multi-layer structure, passes through the first die.

9. The microelectronic assembly of claim 1, wherein the first die is bonded to the second die by metal-to-metal interconnects.

10. The microelectronic assembly of claim 1, wherein one of the first die and the second die includes memory circuitry and another one of the first die and the second die includes processor circuitry.

11. A microelectronic assembly, comprising: a package substrate; a first die, wherein the first die is conductively coupled to the package substrate; a second die; wherein: the first die is at least partially between the second die and the package substrate, the first die is a memory die, the second die is a processor die, the first die is bonded to the second die by first interconnects, wherein the first interconnects are die-to-die interconnects, and the microelectronic assembly includes second interconnects between the second die and the package substrate, wherein the second interconnects do not extend through the first die; and a single mold material over the first die and the second die, the single mold material in contact with a sidewall of the first die, the single mold material having an uppermost surface above an uppermost surface of the second die, and the single mold material on and in contact with the uppermost surface of the second die.

12. The microelectronic assembly of claim 11, wherein the package substrate includes one or more redistribution layers.

13. The microelectronic assembly of claim 11, wherein a footprint of the first die is contained within a footprint of the second die.

14. The microelectronic assembly of claim 11, wherein the first die includes through-substrate vias (TSVs) through a semiconductor substrate of the first die.

15. The microelectronic assembly of claim 11, wherein the first die is conductively coupled to the package substrate by solder.

16. The microelectronic assembly of claim 11, wherein the single mold material is between a portion of the second die and the package substrate, wherein the interconnects extend through the single mold material.

17. The microelectronic assembly of claim 11, wherein a line from the second die to the package substrate, perpendicular to a plane of a layer of dielectric material of the package substrate, passes through the first die.

18. The microelectronic assembly of claim 11, wherein the first die is bonded to the second die by metal-to-metal interconnects.

19. A method of manufacturing a microelectronic assembly, the method including: forming first interconnects between a first die and a second die, wherein the first interconnects are die-to-die interconnects, and wherein the first die has a first surface and an opposing second surface, the first die includes first conductive contacts at its first surface and second conductive contacts at its first surface, the second die has a first surface and an opposing second surface, the second die includes first conductive contacts at its first surface and second conductive contacts at its second surface, and the first interconnects couple the first conductive contacts of the first die with the second

conductive contacts of the second die; after forming the first interconnects, forming second interconnects between the first conductive contacts of the second die and first conductive contacts of a package substrate, and forming third interconnects between the second conductive contacts of the first die and second conductive contacts of the package substrate; and forming a single mold material over the first die and the second die, the single mold material in contact with a sidewall of the first die, the single mold material having an uppermost surface above an uppermost surface of the second die, and the single mold material on and in contact with the uppermost surface of the second die.

20. The method of claim 19, further including: forming an initial package substrate; and forming a recess in the initial package substrate; wherein the first conductive contacts of the package substrate are at a bottom of the recess.

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